Advanced and Quantum Optics

A Comprehensive Study Handout

Part II Physics — University of Cambridge

This handout reconstructs the complete lecture course in a single self-contained document. Every derivation is worked through with the intermediate steps filled in, the physical meaning of each result is spelled out, and the figures and plots from the slides are described in words. It is intended to be read alongside — not instead of — the problem sheets and the textbook (Saleh & Teich, Fundamentals of Photonics*; Gerry & Knight,* Introductory Quantum Optics*).*

Geometric Optics · **Wave & Beam Optics** · **Coherence** · **Fourier Optics** · **Nonlinear Optics** · **Lasers** · **Guided-Wave Optics** · **Resonators** · **Quantum Optics**

The course progresses through four "levels of description" of light: rays, scalar waves, electromagnetic waves, and finally the quantised field.

Contents

1 Course Overview

Executive Summary

Optics can be described at four nested levels of increasing sophistication: **ray (geometric) optics**, valid when the wavelength is negligibly small; **wave optics**, a scalar theory that captures diffraction and interference; **electromagnetic optics**, the full vector Maxwell theory including polarisation; and **quantum optics**, in which the electromagnetic field itself is quantised into photons. This course traverses all four. It begins with ray-transfer matrices and imaging, builds up the scalar wave theory of Gaussian beams, coherence and Fourier optics, then introduces the *nonlinear* response of matter to intense light (harmonic generation, the electro-optic and Kerr effects). It develops the laser from the microscopic physics of stimulated emission, through gain, pumping and resonator feedback, to mode-locking and frequency combs. It treats how light is confined — in dielectric waveguides and fibres, and in optical resonators whose modes are Hermite–Gaussian beams. Finally it quantises the field, introducing Fock and coherent states, the interaction of a quantised field with a two-level atom (the Rabi and Jaynes–Cummings models, optical Bloch equations), and the quantum-statistical signatures of light revealed by photon-correlation experiments (Hanbury Brown–Twiss, Hong–Ou–Mandel). The unifying theme is that each level is a limiting case of the one below it, and that modern photonics — from gravitational-wave detection to quantum computing — draws on all four simultaneously.

How this handout is organised. The material is grouped into eight thematic parts that follow the lecture sequence:

- **Part I** (Lectures 1–2): Geometric optics and ray-transfer matrices.
- **Part II** (Lectures 3–6): Wave optics, coherence, beam optics, microscopy and Fourier optics.
- **Part III** (Lectures 7–9): Nonlinear optics — foundations, the electro-optic and Kerr effects, phase matching and acousto-optics.
- **Part IV** (Lectures 10–12): Lasers — light–matter interaction, gain, oscillation, laser types and pulsed lasers.
- **Part V** (Lectures 13–14): Guided-wave optics — planar and dielectric waveguides, optical fibres.
- **Part VI** (Lectures 15–16): Optical resonators and their Hermite–Gaussian modes.
- **Part VII** (Lectures 17–22): Quantum optics — field quantisation, coherent states, atom–light interaction, cavity QED and photon statistics.

Each part contains the required structure: key concepts, core equations with every variable defined in SI units, step-by-step derivations, worked examples drawn from the problem sheets, and a closing *Common Pitfalls & Takeaways* box.

2 Part I — Geometric Optics (Lectures 1–2)

2.1 Course Overview for this Part

Geometric (ray) optics is the limit of wave optics in which the wavelength $\lambda \to 0$, so that diffraction and interference are absent and light is described purely by *rays* obeying simple geometrical rules. This part establishes the central computational tool of the subject — the 2×2 **ray-transfer (ABCD) matrix** — and applies it to refraction at spherical surfaces, thin and thick lenses, imaging, and compound optical systems such as telescopes.

2.2 Key Concepts & Definitions

- **Optical axis:** the symmetry axis (z axis) of a centred optical system. Rays are characterised by their height y above this axis and their inclination angle θ.
- **Paraxial approximation:** rays make small angles with the optical axis, so $\sin\theta \approx \tan\theta \approx \theta$ and $\cos\theta \approx 1$. All elementary lens theory lives here.
- **Ray-transfer (ABCD) matrix:** a 2×2 matrix M that maps the input ray vector (y_1, θ_1) to the output (y_2, θ_2). Cascaded elements multiply: the matrix of a system is the product of the matrices of its components, written *right to left* in the order light traverses them.
- **Conjugate planes:** an object plane and its image plane. Every point of one maps to a unique point of the other.
- **Aberrations:** the breakdown of perfect imaging for non-paraxial rays (large angles). They are *not* present in the paraxial theory — they are precisely what the paraxial approximation throws away.
- **Cardinal points:** the focal, principal and nodal points that fully characterise a thick lens. For a lens in a single medium the nodal points coincide with the principal points.
- **Collimated beam:** a bundle of parallel rays; its diameter does not change on propagation (an idealisation valid only in ray optics — real beams diverge by diffraction).

2.3 Core Equations

The ray-transfer matrix

$$\begin{pmatrix} y_2 \\ \theta_2 \end{pmatrix} = \begin{pmatrix} A & B \\ C & D \end{pmatrix} \begin{pmatrix} y_1 \\ \theta_1 \end{pmatrix}$$

Here y_1, y_2 are the ray heights (in m) and θ_1, θ_2 the ray angles (in rad) at the input and output planes. The four elements A, B, C, D are dimensionless except B (units $\mathrm{m\,rad^{-1}}$, effectively m) and C (units $\mathrm{rad\,m^{-1}}$, effectively $\mathrm{m^{-1}}$). The elementary building blocks are:

$$\underbrace{\begin{pmatrix} 1 & d \\ 0 & 1 \end{pmatrix}}_{\text{free propagation, distance } d} \qquad \underbrace{\begin{pmatrix} 1 & 0 \\ -1/f & 1 \end{pmatrix}}_{\text{thin lens, focal length } f} \qquad \underbrace{\begin{pmatrix} 1 & 0 \\ \dfrac{n_1 - n_2}{n_2 R} & \dfrac{n_1}{n_2} \end{pmatrix}}_{\text{spherical refracting surface}}$$

where d is a propagation distance (m), f a focal length (m), R the radius of curvature of the surface (m), and n_1, n_2 the refractive indices (dimensionless) before and after the surface. A spherical *mirror* of radius R has the same form as a thin lens with $f = R/2$.

Imaging (Gauss) and Newton equations

$$\frac{1}{s_o} + \frac{1}{s_i} = \frac{1}{f}, \qquad x_o x_i = f^2$$

s_o is the object distance and s_i the image distance from the lens (m); x_o, x_i are the corresponding distances measured from the focal points; f is the focal length (m). The **transverse magnification** is

$$M_T = \frac{y_i}{y_o} = -\frac{s_i}{s_o} = -\frac{x_i}{f} = -\frac{f}{x_o},$$

a dimensionless ratio; a negative value signals an inverted image.

2.4 Important Derivations

Focal length of a spherical mirror: $f = R/2$

Consider a concave mirror of radius R and a paraxial ray travelling parallel to the axis at a small height y. It strikes the mirror at a point where the surface normal (which passes through the centre of curvature C) makes an angle α with the axis, with $\sin\alpha = y/R \approx \alpha$ in the paraxial limit. The law of reflection means the ray is deflected through 2α, so after reflection it crosses the axis at a distance f from the mirror given by $\tan(2\alpha) \approx 2\alpha = y/f$. Eliminating α:

$$2\alpha = \frac{2y}{R} = \frac{y}{f} \quad \Longrightarrow \quad \boxed{f = \frac{R}{2}}.$$

Physical reading: every parallel ray, whatever its height, is bent to the same focal point — but only because we linearised the geometry. Keeping the next term in $\sin\alpha$ would make f depend on y: that height-dependence is *spherical aberration*.

Refraction at a spherical boundary

For a ray passing from medium n_1 to medium n_2 across a spherical interface of radius R, applying Snell's law $n_1 \sin\theta_1 = n_2 \sin\theta_2$ in the paraxial form $n_1\theta_1 = n_2\theta_2$ and doing the small-angle geometry of the triangle formed by the ray, the axis and the surface normal yields

$$\frac{n_1}{z_1} + \frac{n_2}{z_2} = \frac{n_2 - n_1}{R}, \qquad y_2 = -\frac{n_1}{n_2}\frac{z_2}{z_1}\, y_1.$$

The first relation fixes the image distance z_2 *independently of* y_1 — this independence is exactly the statement that all paraxial rays from one object point reconverge to a single image point, i.e. that imaging works. Cascading two such surfaces (the two faces of a lens) and taking the thin-lens limit $d \to 0$ recovers the lensmaker's equation and hence the thin-lens matrix above.

The astronomical telescope

A telescope is two lenses, focal lengths f_1 (objective) and f_2 (eyepiece), separated by $d = f_1 + f_2$ so that the back focal point of the first coincides with the front focal point of the second. The system matrix is the product (right to left: lens 1, gap, lens 2)

$$M = \begin{pmatrix} 1 & 0 \\ -1/f_2 & 1 \end{pmatrix} \begin{pmatrix} 1 & f_1 + f_2 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ -1/f_1 & 1 \end{pmatrix} = \begin{pmatrix} -f_2/f_1 & f_1 + f_2 \\ 0 & -f_1/f_2 \end{pmatrix}.$$

Because $C = 0$, the output angle depends only on the input angle: $\theta_2 = (-f_1/f_2)\,\theta_1$. The system therefore takes parallel input rays to parallel output rays (it is *afocal*) and the **angular**

magnification is

$$M_a = \frac{\theta_2}{\theta_1} = -\frac{f_1}{f_2}.$$

A large objective focal length and a short eyepiece focal length give high magnification; the minus sign means the image is inverted.

Keplerian vs Galilean. If both focal lengths are positive the separation $d = f_1 + f_2$ is positive and the system has a real internal focus — the *Keplerian* telescope, which inverts the image. If the eyepiece is instead a diverging lens ($f_2 < 0$), the separation $d = f_1 + f_2$ is shorter and there is no internal focus — the *Galilean* telescope, which gives an upright image. The same two options apply when the afocal system is used as a beam expander: the Galilean form is more compact and, having no internal focus, avoids air breakdown at high power.

2.5 Example Problem

A worked treatment of cascaded telescopes (achieving a target magnification from a fixed pair of lenses) and of beam-expander design is **Example Sheet 1, Question 2**. The tools needed are all above: the afocal condition $C = 0$, the angular magnification $-f_1/f_2$, and the Keplerian/Galilean distinction just discussed.

Common Pitfalls & Takeaways

1. Matrix ordering. Students routinely multiply the component matrices in the wrong order. Light passes through the *first* element first, but that matrix sits on the *right* of the product (it acts on the input vector first). Always write $M = M_{\text{last}} \cdots M_2 M_1$.

2. "Aberrations are a flaw in the matrices." No — the paraxial matrices are *exact* within the paraxial approximation. Aberrations are the higher-order terms in $\sin\theta$ that the approximation deliberately discards. To study aberrations you must go beyond the linear ray-transfer description entirely.

Takeaway: ray optics is the $\lambda \to 0$ skeleton of all the optics that follows; the ABCD matrix is reused later — unchanged — to propagate *Gaussian beams* via the ABCD law.

3 Part II — Wave, Beam, Coherence & Fourier Optics (Lectures 3–6)

3.1 Course Overview for this Part

When light interacts with features comparable to λ, or is brought to a tight focus, ray optics fails and we need **wave optics** — a scalar theory in which light is a single complex wavefunction obeying the wave equation. This part develops the postulates of wave optics, the all-important **Gaussian beam** (the realistic model of a laser beam), the statistical description of partially coherent light and its measurement by interferometry, the resolution limits of microscopy, and finally **Fourier optics**, in which propagation, diffraction and imaging are all understood as operations on the spatial-frequency spectrum of the field.

3.2 Key Concepts & Definitions

- **Wavefunction $u(\mathbf{r}, t)$:** a real scalar function satisfying the wave equation; it stands for any one Cartesian component of the electric or magnetic field. Polarisation is *not* captured (that needs the vector theory).
- **Complex amplitude $U(\mathbf{r})$:** the time-independent part of the complex wavefunction $U(\mathbf{r}, t) = U(\mathbf{r})e^{i2\pi\nu t}$; it satisfies the Helmholtz equation. The intensity is $I = |U|^2$.

- **Paraxial wave:** a plane wave e^{-ikz} modulated by a slowly varying envelope $A(\mathbf{r})$; the envelope obeys the *paraxial Helmholtz equation.*
- **Gaussian beam:** the fundamental confined solution of the paraxial Helmholtz equation, characterised by a waist radius W_0, a Rayleigh range z_0, an expanding width $W(z)$, a wavefront radius $R(z)$ and the Gouy phase $\zeta(z)$.
- **q-parameter:** the single complex number $q(z) = z + iz_0$ that encodes both $W(z)$ and $R(z)$; it transforms through any ABCD system by the *ABCD law.*
- **Temporal coherence:** the degree to which the field at one time is correlated with the field at a later time; quantified by the *coherence time* τ_c and *coherence length* $\ell_c = c\tau_c$.
- **Complex degree of coherence $g(\tau)$:** the normalised autocorrelation of the field; $|g(\tau)|$ ranges from 1 (fully coherent) to 0 (incoherent).
- **Visibility $\mathcal{V}$:** the fringe contrast $(I_{\max} - I_{\min})/(I_{\max} + I_{\min})$ of an interference pattern; it equals $|g_{12}|$ for equal-intensity beams.
- **Numerical aperture, Abbe limit:** $\mathrm{NA} = n \sin\theta$; the smallest resolvable feature is $d = \lambda/(2\,\mathrm{NA})$.
- **Transfer function / impulse response:** in Fourier optics, free space acts as a linear shift-invariant system; $H(\nu_x, \nu_y)$ multiplies the spatial spectrum, $h(x, y)$ convolves the field.
- **Fraunhofer (far-field) diffraction:** the diffraction pattern is the squared modulus of the Fourier transform of the aperture function; a lens brings this far field into its focal plane.

3.3 Core Equations

Wave equation and Helmholtz equation

$$\nabla^2 u - \frac{1}{c^2}\frac{\partial^2 u}{\partial t^2} = 0, \qquad \nabla^2 U + k^2 U = 0, \qquad k = \frac{2\pi\nu}{c} = \frac{2\pi}{\lambda}.$$

u is the real wavefunction; $c = c_0/n$ is the speed of light in the medium ($\mathrm{m\,s^{-1}}$); U is the complex amplitude; k is the wavenumber ($\mathrm{rad\,m^{-1}}$); ν the frequency (Hz); λ the wavelength in the medium (m). The optical intensity is $I(\mathbf{r}) = |U(\mathbf{r})|^2$ (units $\mathrm{W\,m^{-2}}$).

The Gaussian beam

$$U(\mathbf{r}) = A_0 \frac{W_0}{W(z)} \exp\left[-\frac{\rho^2}{W^2(z)}\right] \exp\left[-ikz - ik\frac{\rho^2}{2R(z)} + i\zeta(z)\right]$$

with $\rho^2 = x^2 + y^2$ the transverse radius (m) and

$$W(z) = W_0\sqrt{1 + (z/z_0)^2}, \quad R(z) = z\left[1 + (z_0/z)^2\right], \quad \zeta(z) = \arctan(z/z_0), \quad W_0 = \sqrt{\frac{\lambda z_0}{\pi}}.$$

Variables: W_0 is the waist radius (m); $z_0 = \pi W_0^2/\lambda$ the Rayleigh range (m); $W(z)$ the $1/e^2$-intensity radius at position z (m); $R(z)$ the wavefront radius of curvature (m); $\zeta(z)$ the Gouy phase (rad); A_0 a constant amplitude. Useful derived quantities:

$$\text{divergence half-angle} \quad \theta_0 = \frac{\lambda}{\pi W_0}, \qquad \text{depth of focus} \quad 2z_0 = \frac{2\pi W_0^2}{\lambda}.$$

The q-parameter and the ABCD law

$$\frac{1}{q(z)} = \frac{1}{R(z)} - \mathrm{i}\,\frac{\lambda}{\pi W^2(z)}, \qquad q(z) = z + \mathrm{i}z_0, \qquad q_2 = \frac{Aq_1 + B}{Cq_1 + D}.$$

The same A, B, C, D matrix that propagates rays propagates a Gaussian beam through the transformation of q shown on the right.

Coherence

$$G(\tau) = \langle U^*(t)\, U(t+\tau)\rangle\,, \qquad g(\tau) = \frac{G(\tau)}{G(0)}, \qquad \tau_c = \int_{-\infty}^{\infty} |g(\tau)|^2\,\mathrm{d}\tau, \qquad \Delta\nu_c = \frac{1}{\tau_c}.$$

$G(\tau)$ is the temporal coherence function; $g(\tau)$ its normalised version (dimensionless); τ_c the coherence time (s); $\ell_c = c\tau_c$ the coherence length (m); $\Delta\nu_c$ the spectral width (Hz). The autocorrelation and the power spectral density $S(\nu)$ form a Fourier pair (Wiener–Khinchin theorem): $S(\nu) = \int G(\tau)\mathrm{e}^{-\mathrm{i}2\pi\nu\tau}\mathrm{d}\tau$.

Interference and the Airy pattern

$$I = I_1 + I_2 + 2\sqrt{I_1 I_2}\,|g_{12}|\cos\varphi, \qquad \mathcal{V} = \frac{2\sqrt{I_1 I_2}}{I_1 + I_2}\,|g_{12}|.$$

$$I(\rho) = I_0\left[\frac{2J_1(\pi D\rho/\lambda d)}{\pi D\rho/\lambda d}\right]^2, \qquad \text{Airy disk radius } \rho_A = 1.22\,\frac{\lambda d}{D}.$$

I_1, I_2 are the two beam intensities; $\varphi = \arg g_{12}$ the phase; J_1 the first-order Bessel function; D the aperture diameter (m); d the propagation distance (m). The Abbe resolution limit and Rayleigh criterion are $d_{\min} = \lambda/(2\,\mathrm{NA})$ and $\Delta\ell \approx 1.22\,\lambda f/D$.

3.4 Important Derivations

From the wave equation to the Helmholtz equation

Insert a monochromatic complex wavefunction $U(\mathbf{r}, t) = U(\mathbf{r})\,\mathrm{e}^{\mathrm{i}2\pi\nu t}$ into the wave equation. The time derivative brings down a factor $(\mathrm{i}2\pi\nu)^2 = -4\pi^2\nu^2$:

$$\nabla^2 U\,\mathrm{e}^{\mathrm{i}2\pi\nu t} - \frac{1}{c^2}(-4\pi^2\nu^2)\,U\,\mathrm{e}^{\mathrm{i}2\pi\nu t} = 0.$$

Cancelling the common exponential and identifying $k = 2\pi\nu/c$ gives the Helmholtz equation $\nabla^2 U + k^2 U = 0$. The physical content: a single temporal frequency has been separated out, leaving a purely spatial equation for the field pattern.

The paraxial Helmholtz equation

Write a paraxial wave as a plane-wave carrier times a slow envelope, $U(\mathbf{r}) = A(\mathbf{r})\,\mathrm{e}^{-\mathrm{i}kz}$. Substituting into the Helmholtz equation and differentiating,

$$\nabla_T^2 A\,\mathrm{e}^{-\mathrm{i}kz} + \frac{\partial^2 A}{\partial z^2}\mathrm{e}^{-\mathrm{i}kz} - 2\mathrm{i}k\frac{\partial A}{\partial z}\mathrm{e}^{-\mathrm{i}kz} - k^2 A\,\mathrm{e}^{-\mathrm{i}kz} + k^2 A\,\mathrm{e}^{-\mathrm{i}kz} = 0,$$

where $\nabla_T^2 = \partial_x^2 + \partial_y^2$. The last two terms cancel. The *slowly varying envelope approximation* says A changes little over one wavelength, so $|\partial_z^2 A| \ll |k\,\partial_z A|$ and we drop $\partial_z^2 A$. The result is the

Paraxial Helmholtz equation

$$\nabla_T^2 A - 2\mathrm{i}k\,\frac{\partial A}{\partial z} = 0.$$

This is mathematically a 2D Schrödinger equation with z playing the role of time — the Gaussian beam is its "ground state".

Properties of the Gaussian beam from the q-parameter

The paraboloidal wave $A = (A_1/q)\exp[-\mathrm{i}k\rho^2/2q]$ with $q(z) = z + \mathrm{i}z_0$ solves the paraxial equation. Splitting $1/q$ into real and imaginary parts,

$$\frac{1}{q} = \frac{1}{z+\mathrm{i}z_0} = \frac{z-\mathrm{i}z_0}{z^2+z_0^2} = \underbrace{\frac{z}{z^2+z_0^2}}_{\equiv 1/R(z)} - \mathrm{i}\underbrace{\frac{z_0}{z^2+z_0^2}}_{\equiv \lambda/\pi W^2(z)}.$$

The real part defines the wavefront curvature; inverting it gives $R(z) = z[1+(z_0/z)^2]$. The imaginary part defines the width; using $z_0 = \pi W_0^2/\lambda$ and simplifying gives $W(z) = W_0\sqrt{1+(z/z_0)^2}$. Substituting back and collecting the prefactor's phase produces the Gouy phase $\zeta(z) = \arctan(z/z_0)$. *Physical reading:* a single complex number q carries the entire beam geometry, and because q transforms by the ABCD law, designing laser optics is again "just matrix multiplication".

Beam power is independent of z

Integrate the Gaussian intensity $I(\rho,z) = I_0[W_0/W(z)]^2\exp[-2\rho^2/W^2(z)]$ over a transverse plane:

$$P = \int_0^\infty I(\rho,z)\,2\pi\rho\,\mathrm{d}\rho = I_0\frac{W_0^2}{W^2}\,2\pi\int_0^\infty \mathrm{e}^{-2\rho^2/W^2}\rho\,\mathrm{d}\rho = I_0\frac{W_0^2}{W^2}\,2\pi\cdot\frac{W^2}{4} = \tfrac{1}{2}I_0\,\pi W_0^2.$$

The $W(z)$ dependence cancels: as the beam expands its peak intensity falls exactly fast enough to conserve power, as it must for a source-free field.

Coherence time of an exponential and a Gaussian envelope

(Problem Set 1, Q4.) For $g(\tau) = \exp(-|\tau|/\tau_c)$,

$$\int_{-\infty}^{\infty}|g(\tau)|^2\mathrm{d}\tau = 2\int_0^\infty \mathrm{e}^{-2\tau/\tau_c}\mathrm{d}\tau = 2\cdot\frac{\tau_c}{2} = \tau_c,$$

consistent with the definition. For the Gaussian $g(\tau) = \exp(-\pi\tau^2/2\tau_c^2)$,

$$\int_{-\infty}^{\infty}\mathrm{e}^{-\pi\tau^2/\tau_c^2}\mathrm{d}\tau = \sqrt{\frac{\pi}{\pi/\tau_c^2}} = \tau_c,$$

again consistent. As τ runs from 0 to τ_c, $|g|$ falls to $\mathrm{e}^{-1}\approx 0.37$ (exponential) or $\mathrm{e}^{-\pi/2}\approx 0.21$ (Gaussian): the coherence time is the natural "memory time" of the fluctuations.

The interference law for partially coherent light

Superpose two partially coherent fields and take the ensemble average:

$$I = \langle|U_1+U_2|^2\rangle = \langle|U_1|^2\rangle + \langle|U_2|^2\rangle + \langle U_1^*U_2\rangle + \langle U_1U_2^*\rangle = I_1 + I_2 + 2\,\mathrm{Re}\{G_{12}\}.$$

Writing the normalised cross-correlation $g_{12} = G_{12}/\sqrt{I_1I_2} = |g_{12}|\mathrm{e}^{\mathrm{i}\varphi}$ gives the interference law $I = I_1 + I_2 + 2\sqrt{I_1I_2}\,|g_{12}|\cos\varphi$. The fringe maxima and minima are $I_{\max/\min} = I_1 + I_2 \pm 2\sqrt{I_1I_2}\,|g_{12}|$, so the visibility is $\mathcal{V} = 2\sqrt{I_1I_2}\,|g_{12}|/(I_1+I_2)$. *Measuring fringe contrast measures $|g|$ directly.*

Free-space propagation as a Fourier operation

Decompose the input field at $z = 0$ into plane waves: $f(x, y) = \iint F(\nu_x, \nu_y)\, \mathrm{e}^{-\mathrm{i}2\pi(\nu_x x+\nu_y y)} \mathrm{d}\nu_x \mathrm{d}\nu_y$. Each plane-wave component propagates a distance d by acquiring the phase $\mathrm{e}^{-\mathrm{i}k_z d}$ with $k_z = 2\pi\sqrt{\lambda^{-2} - \nu_x^2 - \nu_y^2}$. Hence the **transfer function of free space** is

$$H(\nu_x, \nu_y) = \exp\left[-\mathrm{i}2\pi d\sqrt{\lambda^{-2} - \nu_x^2 - \nu_y^2}\right].$$

For $\nu_x^2 + \nu_y^2 > \lambda^{-2}$ the square root is imaginary and H becomes a real *decaying* exponential: these are **evanescent waves**, and the fact that they decay is precisely why features finer than λ cannot propagate to the far field. In the paraxial (Fresnel) limit, expanding the square root gives $H \approx H_0 \exp[\mathrm{i}\pi\lambda d(\nu_x^2 + \nu_y^2)]$ — a parabolic phase. Taking the far field (Fraunhofer limit, Fresnel number $N_F = a^2/\lambda d \ll 1$) leaves

$$g(x, y) \approx h_0\, F\left(\frac{x}{\lambda d}, \frac{y}{\lambda d}\right),$$

i.e. *the field in the far field is the Fourier transform of the input field.* A lens performs this transform in its focal plane without needing the large distance d.

3.5 Diagram Analysis

Several slides plot the Gaussian-beam quantities. The plot of $W(z)$ versus z is a hyperbola: flat near the waist, asymptoting to straight lines of slope θ_0 far away. The plot of on-axis intensity $I(0, z) = I_0/[1 + (z/z_0)^2]$ is a Lorentzian in z, falling to half its peak at $z = \pm z_0$ — this defines the depth of focus $2z_0$. The diagrams of the $4f$ system with a mask in the Fourier plane show that blocking the centre of the spectrum gives a high-pass (edge- enhanced) image, while a small central aperture gives a low-pass (blurred) image. The interferograms shown for OCT consist of separate fringe bursts, one per reflecting layer, each of temporal width τ_c centred on that layer's round-trip delay.

3.6 Example Problems

Two problem-sheet questions consolidate this material.

Focusing and collimating a Gaussian beam — Example Sheet 1, Question 8. A thin lens transforms an incident Gaussian beam into a new Gaussian beam. The relevant result (worth knowing) is that the new waist radius is $W_0' = MW_0$ with

$$M = \frac{M_r}{\sqrt{1 + r^2}}, \qquad M_r = \left|\frac{f}{z - f}\right|, \qquad r = \frac{z_0}{z - f},$$

where z is the input waist-to-lens distance and the new waist sits at $(z' - f) = M^2(z - f)$. "Best collimation" means pushing the output waist as far from the lens as possible, which occurs for $z = f + z_0$; the question asks for M, W_0' and z' in a specific numerical case.

Optical Doppler radar / heterodyne detection — Example Sheet 1, Question 3. A wave returning from a target moving at line-of-sight speed v is Doppler-shifted by $\Delta\nu = \pm(2v/c)\nu$. Superposed on an unshifted reference, the detected intensity is $I(t) = 2I_0[1 + \cos(2\pi\, \Delta\nu\, t)]$ — a slow beat at the difference frequency. The key idea is that mixing the signal with a reference converts an immeasurably fast *optical* frequency shift into a slow, electronically measurable beat note; this is the principle behind all heterodyne (coherent) detection, including LIDAR and the moving-mirror Michelson interferometer.

Common Pitfalls & Takeaways

1. "$W(z)$ is the beam edge." A Gaussian beam has no sharp edge; $W(z)$ is the radius at which the *intensity* has fallen to $1/e^2$ (the *amplitude* to $1/e$). About 86% of the power lies within $\rho < W$.
2. Confusing the coherence *time* with the pulse duration, or τ_c with $1/\nu$. The coherence time is set by the spectral *width* $\Delta\nu_c$ (via $\tau_c \sim 1/\Delta\nu_c$), not by the optical frequency ν. A narrow-linewidth CW laser can have τ_c of microseconds while oscillating at $\sim 10^{14}$ Hz.
3. Forgetting that diffraction sets a hard floor. You cannot simultaneously make W_0 small (tight focus) *and* θ_0 small (low divergence): their product is fixed at λ/π. Squeezing the spot necessarily increases the divergence and shortens the depth of focus.
Takeaway: Fourier optics unifies propagation, diffraction and imaging — they are all the *same* linear operation viewed in the spatial-frequency domain. Evanescent waves carry the sub-wavelength information but do not survive to the far field, which is why classical microscopes are diffraction-limited.

4 Part III — Nonlinear Optics (Lectures 7–9)

4.1 Course Overview for this Part

At ordinary intensities the polarisation of a medium is proportional to the applied field, $P = \epsilon_0 \chi E$, and optics is linear: beams pass through one another untouched and the refractive index is fixed. At the enormous field strengths available from lasers, the next terms in the expansion $P = \epsilon_0(\chi E + \chi^{(2)}E^2 + \chi^{(3)}E^3 + \cdots)$ become significant. These **nonlinear** terms let light of one frequency generate light at new frequencies (harmonic generation, sum- and difference-frequency mixing), let a DC field control the refractive index (the Pockels effect), and let light control its own propagation (the Kerr effect, self-focusing, solitons). The whole subject hinges on two bookkeeping rules — conservation of energy ($\sum \omega$) and momentum ($\sum \mathbf{k}$, i.e. *phase matching*) — and on the crystal symmetry that decides which $\chi^{(n)}$ are allowed to be non-zero.

4.2 Key Concepts & Definitions

- **Anharmonic oscillator model:** the electron is bound in a potential with cubic/quartic corrections; the cubic term produces $\chi^{(2)}$, the quartic term $\chi^{(3)}$.
- **Second-harmonic generation (SHG):** two photons at ω combine into one at 2ω — the $\cos(2\omega t)$ term in $P^{(2)}$.
- **Optical rectification:** the DC ($\omega = 0$) term in $P^{(2)}$; an intense optical beam induces a static polarisation.
- **Three-wave mixing:** the general $\chi^{(2)}$ process $\omega_1 + \omega_2 = \omega_3$, encompassing sum-frequency, difference- frequency and parametric generation.
- **Phase matching:** the momentum-conservation condition $\mathbf{k}_1 + \mathbf{k}_2 = \mathbf{k}_3$; without it the generated wave slips out of phase with its driving polarisation and the conversion efficiency collapses after one *coherence length* $L_c = 2\pi/|\Delta k|$.
- **Birefringent phase matching:** using the difference between ordinary and extraordinary refractive indices in an anisotropic crystal to cancel material dispersion and satisfy phase matching.
- **Quasi-phase matching:** periodically reversing the sign of $\chi^{(2)}$ (periodic poling) with period $\Lambda = L_c$ to reset the phase error every coherence length.

- **Pockels effect:** a refractive index that varies *linearly* with an applied DC field, $n(E_0) = n_0 + \frac{1}{2}(\chi^{(2)}/n_0)E_0$; only in non-centrosymmetric crystals.
- **Kerr effect:** a refractive index that varies with optical *intensity*, $n(I) = n_0 + n_2 I$; present even in centrosymmetric media.
- **Self-phase modulation / self-focusing:** the Kerr effect acting on a beam's own intensity profile — temporally it chirps a pulse, spatially it focuses a beam.
- **Soliton:** a pulse (or beam) in which Kerr self-focusing exactly balances dispersion (or diffraction), so the shape propagates unchanged; described by the nonlinear Schrödinger equation.
- **Acousto-optic effect:** a travelling sound wave is a moving strain grating that Bragg-diffracts light, shifting its frequency by the acoustic frequency; the basis of acousto-optic modulators (AOMs).

4.3 Core Equations

The nonlinear polarisation

$$P = \epsilon_0\Big(\chi^{(1)}E + \chi^{(2)}E^2 + \chi^{(3)}E^3 + \cdots\Big)$$

P is the polarisation ($\mathrm{C\,m^{-2}}$); E the electric field ($\mathrm{V\,m^{-1}}$); ϵ_0 the vacuum permittivity ($\mathrm{F\,m^{-1}}$); $\chi^{(1)}$ is dimensionless; $\chi^{(2)}$ has units $\mathrm{m\,V^{-1}}$; $\chi^{(3)}$ has units $\mathrm{m^2\,V^{-2}}$. In anisotropic crystals each $\chi^{(n)}$ is a tensor; the contracted second-order coefficient is written d_{iJ} with $\epsilon_0\chi^{(2)}_{ijk} = 2d_{iJ}$.

Nonlinear wave equation

$$\nabla^2 E - \frac{1}{c^2}\frac{\partial^2 E}{\partial t^2} = \mu_0\frac{\partial^2 P_{\mathrm{NL}}}{\partial t^2}$$

The linear part of P is absorbed into $c = c_0/n$; the nonlinear part P_{NL} acts as a *source term* — an oscillating nonlinear polarisation radiates new frequencies. μ_0 is the vacuum permeability ($\mathrm{H\,m^{-1}}$).

Conservation laws / phase matching

$$\omega_1 + \omega_2 = \omega_3 \quad \text{(energy)}, \qquad \mathbf{k}_1 + \mathbf{k}_2 = \mathbf{k}_3 \quad \text{(momentum / phase matching)}.$$

ω_i are angular frequencies ($\mathrm{rad\,s^{-1}}$); $\mathbf{k}_i$ are wavevectors ($\mathrm{rad\,m^{-1}}$). The coherence length for a residual mismatch Δk is $L_c = 2\pi/|\Delta k|$ (m).

Electro-optic and Kerr index changes

$$n(E_0) \approx n_0 + \frac{\chi^{(2)}(\omega;\omega,0)}{2n_0}E_0 \quad \text{(Pockels)}, \qquad n(I) \approx n_0 + n_2 I, \quad n_2 = \frac{\chi^{(3)}(\omega;\omega,\omega,-\omega)}{2n_0^2\epsilon_0 c} \quad \text{(Kerr)}.$$

n_0 is the linear refractive index; E_0 the applied DC field ($\mathrm{V\,m^{-1}}$); I the optical intensity ($\mathrm{W\,m^{-2}}$); n_2 the nonlinear index ($\mathrm{m^2\,W^{-1}}$, typically $\sim 10^{-20}\ \mathrm{m^2\,W^{-1}}$ for glass).

Acousto-optics: the Bragg condition

$$\sin\theta_B = \frac{\lambda}{2\Lambda}, \qquad \mathbf{k}_r = \mathbf{k} + \mathbf{q}, \qquad \nu_r = \nu + f.$$

θ_B is the Bragg angle; λ the optical wavelength in the medium (m); Λ the acoustic wavelength (m); $\mathbf{q}$ the acoustic wavevector; f the acoustic frequency (Hz); ν_r the diffracted (frequency-shifted) optical frequency. The strain–index relation is $\Delta n = -\frac{1}{2}pn^3 S$ with p the (dimensionless) elasto-optic coefficient and S the strain.

4.4 Important Derivations

Where SHG and rectification come from

Model the bound electron in a weakly anharmonic potential $V = \frac{1}{2}k_0 x^2 + \frac{1}{3}k_1 x^3 + \cdots$ with $k_1 \ll k_0$. The restoring force is $F = -k_0 x - k_1 x^2 - \cdots$. Solving perturbatively for the displacement under an applied force F_a gives a quadratic correction, and hence the polarisation acquires a term $\propto E^2$. Now drive with $E(t) = E\cos(\omega t)$ and use the identity $\cos^2(\omega t) = \frac{1}{2}[1 + \cos(2\omega t)]$:

$$P^{(2)}(t) \propto E^2 \cos^2(\omega t) = \tfrac{1}{2}E^2 \Big(\underbrace{1}_{\text{rectification}} + \underbrace{\cos(2\omega t)}_{\text{2nd harmonic}} \Big).$$

The constant term is a static polarisation (*optical rectification*); the $\cos(2\omega t)$ term is an oscillating polarisation that radiates at 2ω (*second-harmonic generation*). Crucially, $k_1 = 0$ in any centrosymmetric medium — the potential must be *asymmetric* for $\chi^{(2)}$ to exist.

Why centrosymmetric media have $\chi^{(2)} = 0$

In a centrosymmetric medium, inverting all coordinates is a symmetry, so reversing the field must reverse the polarisation: $E \to -E \Rightarrow P \to -P$. But the term $\chi^{(2)}E^2$ is unchanged under $E \to -E$. The only way to satisfy $-\chi^{(2)}E^2 = \chi^{(2)}(-E)^2 = \chi^{(2)}E^2$ for all E is $\chi^{(2)} = 0$. Third-order terms $\chi^{(3)}E^3$ *are* odd under $E \to -E$, so $\chi^{(3)}$ survives in *any* medium.

The Pockels effect is rectification run backwards

Consider a $\chi^{(2)}$ process with one optical field at ω and one DC field at 0: $P^{(2)}(\omega) = \chi^{(2)}(\omega;\omega,0)E(\omega)E_0$. The total displacement field is

$$D = \epsilon_0 \Big[1 + \chi^{(1)} + \chi^{(2)}(\omega;\omega,0)E_0\Big] E,$$

so the medium behaves *linearly* but with an effective index

$$n(E_0) = \sqrt{1 + \chi^{(1)} + \chi^{(2)}E_0} \approx n_0 + \frac{\chi^{(2)}}{2n_0}E_0,$$

expanding the square root because the nonlinear term is small. By the permutation symmetry of $\chi^{(2)}$ in non-dispersive media, $\chi^{(2)}(\omega;\omega,0) = \chi^{(2)}(0;\omega,-\omega)$ — the coefficient governing the Pockels effect is literally the same one that governs optical rectification.

The optical Kerr effect

Now take the third-order process $\chi^{(3)}(\omega;\omega,\omega,-\omega)$ with a single optical field. Since $E(\omega)E(\omega)E^*(\omega) = E(\omega)|E(\omega)|^2$ and $I = n_0\epsilon_0 c|E|^2$, the third-order polarisation is $P^{(3)}(\omega) = [\chi^{(3)}/(n_0\epsilon_0 c)]\,E(\omega)\,I$. The effective index then depends on the beam's *own* intensity:

$$n(I) = \sqrt{1 + \chi^{(1)} + \frac{\chi^{(3)}}{n_0\epsilon_0 c}I} \approx n_0 + \underbrace{\frac{\chi^{(3)}}{2n_0^2\epsilon_0 c}}_{\equiv n_2} I.$$

Self-phase modulation and the chirp

A pulse travelling through a Kerr medium accumulates the phase $\varphi(t) = \omega_0 t - k_0 z[n + n_2 I(t)]$. The *instantaneous frequency* is the time derivative:

$$\omega(t) = \frac{\partial \varphi}{\partial t} = \omega_0 - k_0 z\, n_2 \, \frac{\partial I(t)}{\partial t}.$$

On the rising edge of the pulse $\partial I/\partial t > 0$, so the frequency is lowered (red-shifted); on the falling edge it is raised. The pulse acquires a frequency sweep — a *chirp*. Spatially, the same effect makes the centre of a beam (high I) experience a higher index than its wings, so the medium acts as a positive lens: the beam *self-focuses*.

Phase mismatch and the coherence length

The generated intensity is the squared integral of the source amplitude over the interaction length, with the residual mismatch Δk appearing as a phase factor:

$$I_3 \propto \left| \int_0^L \mathrm{e}^{\mathrm{i}\Delta k\, z}\, \mathrm{d}z \right|^2 = L^2\, \mathrm{sinc}^2\left(\frac{\Delta k\, L}{2}\right).$$

This is maximal at $\Delta k = 0$ (perfect phase matching) and falls to its first zero when $\Delta k L/2 = \pi$, i.e. at the **coherence length** $L_c = 2\pi/|\Delta k|$. Beyond L_c the generated wave is 180° out of phase with the polarisation driving it and energy flows *back* into the fundamental. Quasi-phase matching defeats this by flipping the sign of d every L_c (periodic poling), so the phase error is reset before it can reverse the power flow.

4.5 Diagram Analysis

The slides contrast the harmonic and anharmonic potentials: the harmonic parabola gives a sinusoidal polarisation response, whereas the asymmetric (cubic) potential distorts the response so that it contains a DC offset and a doubled-frequency component — a direct visual of where $\chi^{(2)}$ comes from. The $\mathrm{sinc}^2(\Delta k L/2)$ curve shows the conversion efficiency: a sharp central peak at $\Delta k = 0$ with rapidly decaying side-lobes, emphasising how stringent phase matching becomes for long crystals. The soliton figures juxtapose a pulse that broadens under pure dispersion against one that holds its shape when the Kerr nonlinearity is added — the two effects cancelling. The index-surface diagrams (a circle for the ordinary wave, an ellipse for the extraordinary wave) show graphically how rotating the propagation direction relative to the optic axis tunes the extraordinary index until $n_e(\theta, \omega) = n_o(2\omega)$ and phase matching is achieved.

4.6 Example Problems

When do nonlinear terms matter? — Example Sheet 2, Question 1. This question asks at what intensity the $\chi^{(3)}$ term reaches 1% of the linear term in carbon disulphide. The method is to form the ratio $|\chi^{(3)} E^3|/|\chi^{(1)} E| = (\chi^{(3)}/\chi^{(1)}) E^2$, set it to 0.01, and convert field to intensity with $I = \frac{1}{2} n \epsilon_0 c_0 E^2$. The answer comes out of order $10^{14}\ \mathrm{W\,cm^{-2}}$ — the headline point being that nonlinear optics needs focused, pulsed laser intensities many orders of magnitude above sunlight, which is why it is a laser-era subject.

Phase matching for second-harmonic generation — Example Sheet 2, Question 5. This asks you to show that collinear Type-I SHG requires $n(2\omega) = n(\omega)$, and to explain why this fails in an isotropic medium. Both points are derived in full in the *Phase mismatch and the coherence length* derivation above and the *birefringent phase matching* concept: normal dispersion forces $n(2\omega) > n(\omega)$ in an isotropic medium, so the equality can be met only by using the birefringence of an anisotropic crystal as the extra tuning knob.

Common Pitfalls & Takeaways

1. "More crystal length always means more harmonic." Only if you are phase matched. With a finite Δk, the harmonic intensity oscillates with length as $\mathrm{sinc}^2(\Delta k L/2)$ — past one coherence length the conversion actually *decreases*.
2. Confusing the Pockels and Kerr effects. Pockels is *linear* in an applied DC field and exists only in non-centrosymmetric crystals; the optical Kerr effect is linear in optical *intensity* and exists in any material. Check which field the index depends on.
3. Treating $\chi^{(2)}$ as a single number. In a real crystal it is a tensor: which component you use depends on the polarisations of the three waves and their orientation relative to the crystal axes. The contracted d_{iJ} table exists precisely to keep this bookkeeping straight.
Takeaway: nonlinear optics is governed by two conservation laws (energy fixes the frequencies, momentum/phase-matching fixes whether the process is efficient) and one symmetry rule (centrosymmetry kills all even-order χ). Master those three statements and the zoo of processes becomes orderly.

5 Part IV — Lasers (Lectures 10–12)

5.1 Course Overview for this Part

A laser is built from three ingredients: an **amplifier** (a medium with population inversion that amplifies light by stimulated emission), a **pump** (an energy source that maintains the inversion), and **feedback** (an optical resonator that returns the light through the amplifier). This part derives the laser from the bottom up: the quantum laws governing absorption, spontaneous emission and stimulated emission; the gain coefficient and its frequency dependence (the lineshape); the rate equations describing how pumping builds inversion and how strong fields saturate it; the threshold and steady-state output of the oscillator; and finally the practical landscape — laser types, linewidth limits, pulsed operation by Q-switching and mode-locking, and the optical frequency comb.

5.2 Key Concepts & Definitions

- **Absorption / stimulated emission / spontaneous emission:** the three ways an atom and a radiation mode exchange a photon. Absorption and stimulated emission require a photon already present; spontaneous emission does not.
- **Transition cross section $\sigma(\nu)$:** the effective atomic area for a photon-induced transition; its area is the transition strength S, its shape the normalised *lineshape function* $g(\nu)$.
- **Population inversion $N = N_2 - N_1$:** the excess of upper- over lower-level population. $N > 0$ gives gain; $N < 0$ (thermal equilibrium) gives absorption; $N = 0$ is transparent.
- **Gain coefficient $\gamma(\nu)$:** the fractional growth of intensity per unit length, $\gamma(\nu) = N\sigma(\nu)$.
- **Pumping and rate equations:** the differential equations balancing pump, decay and stimulated transitions that determine the steady-state inversion N_0.
- **Gain saturation:** as the intracavity field grows, it depletes the inversion, so $\gamma = \gamma_0/(1+\phi/\phi_s)$ — the gain falls with increasing photon flux ϕ.
- **Resonator loss α_r and photon lifetime τ_p:** the total round-trip loss per unit length and the corresponding $1/e$ decay time of stored photons, $\tau_p = 1/(c\alpha_r)$.
- **Threshold:** oscillation begins when the small-signal gain exceeds the loss, $\gamma_0(\nu) > \alpha_r$, equivalently $N_0 > N_t$.

- **Gain clamping:** above threshold the saturated gain is pinned at the loss value; extra pump power goes into output photons, not into more gain.
- **Homogeneous vs inhomogeneous broadening:** whether all atoms share the same lineshape (homogeneous — gives single-mode operation via uniform saturation) or different atoms have shifted lines (inhomogeneous — gives *spectral hole burning* and multimode operation).
- **Schawlow–Townes limit:** the fundamental quantum lower bound on laser linewidth, set by phase diffusion from spontaneous emission, $\Delta\nu_l \sim 1/(\tau_p N_{\mathrm{ph}})$.
- **Q-switching:** pulsing by suddenly lowering the resonator loss after storing energy in the inverted population.
- **Mode-locking:** pulsing by locking the phases of many longitudinal modes so they interfere into a periodic train of short, intense pulses.
- **Optical frequency comb:** the spectrum of a mode-locked laser — a set of perfectly even "teeth" $\nu_q = q\nu_F + \nu_i$, a ruler for measuring optical frequencies.

5.3 Core Equations

Transition rates and the lineshape

$$W_i = \phi\,\sigma(\nu), \qquad \sigma(\nu) = S\,g(\nu), \qquad \int_0^\infty g(\nu)\,\mathrm{d}\nu = 1, \qquad \bar{\sigma}(\nu) = \frac{\lambda^2}{8\pi t_{sp}}\,g(\nu).$$

W_i is the stimulated transition probability density (s^{-1}); ϕ the photon-flux density ($\mathrm{cm}^{-2}\,\mathrm{s}^{-1}$); $\sigma(\nu)$ the transition cross section (cm^2); S the transition strength ($\mathrm{cm}^2\,\mathrm{Hz}$); $g(\nu)$ the lineshape (Hz^{-1}); t_{sp} the spontaneous-emission lifetime (s); λ the wavelength in the medium (m). The last relation is the *Füchtbauer–Ladenburg equation*.

Gain coefficient and its growth

$$\gamma(\nu) = N\,\sigma(\nu) = N\,\frac{\lambda^2}{8\pi t_{sp}}\,g(\nu), \qquad \frac{\mathrm{d}\phi}{\mathrm{d}z} = \gamma(\nu)\,\phi \quad\Rightarrow\quad \phi(z) = \phi(0)\,\mathrm{e}^{\gamma(\nu)z}.$$

$\gamma(\nu)$ is the gain coefficient (m^{-1}); $N = N_2 - N_1$ the population-density difference (m^{-3}). When $N < 0$ this becomes the attenuation law $\phi(z) = \phi(0)\mathrm{e}^{-\alpha(\nu)z}$ with $\alpha = -\gamma$.

Steady-state inversion, saturation, threshold

$$N_0 \approx R_2 t_{sp} + R_1\tau_1, \qquad N = \frac{N_0}{1+\tau_s W_i}, \qquad \gamma(\nu) = \frac{\gamma_0(\nu)}{1+\phi/\phi_s(\nu)}, \qquad N_t = \frac{\alpha_r}{\sigma(\nu)} = \frac{1}{c\tau_p\sigma(\nu)}.$$

R_1, R_2 are pump rates into levels 1, 2 ($\mathrm{m}^{-3}\,\mathrm{s}^{-1}$); τ_1, τ_2, t_{sp} are level lifetimes (s); τ_s the saturation time constant (s); $\phi_s = (\tau_s\sigma)^{-1}$ the saturation photon flux; γ_0 the small-signal (unsaturated) gain (m^{-1}); α_r the total loss coefficient (m^{-1}); τ_p the photon lifetime (s); N_t the threshold inversion (m^{-3}).

Resonator: modes, loss, output

$$\nu_q = q\,\nu_F, \quad \nu_F = \frac{c}{2d}, \qquad \alpha_r = \alpha_s + \frac{1}{2d}\ln\frac{1}{R_1R_2}, \qquad \phi = \phi_s(\nu)\left(\frac{N_0}{N_t} - 1\right) \text{ for } N_0 > N_t.$$

ν_q are the longitudinal mode frequencies; ν_F the free spectral range (Hz); d the resonator length (m); α_s the distributed (absorption/scattering) loss (m^{-1}); R_1, R_2 the mirror reflectances (dimensionless). Above threshold the internal photon flux rises linearly with the pump.

Mode-locking

$$I(t) = |A|^2 \frac{\sin^2[M\pi(t-z/c)/T_F]}{\sin^2[\pi(t-z/c)/T_F]}, \qquad T_F = \frac{1}{\nu_F} = \frac{2d}{c}, \qquad \tau_{\text{pulse}} = \frac{T_F}{M} = \frac{1}{\Delta\nu}, \qquad I_p = M\bar{I}.$$

M is the number of locked modes; T_F the pulse repetition period (s); τ_{pulse} the pulse duration (s); $\Delta\nu = M\nu_F$ the locked bandwidth (Hz); $\bar{I}, I_p$ the mean and peak intensities. The frequency-comb teeth are $\nu_q = q\nu_F + \nu_i$ with offset ν_i.

5.4 Important Derivations

A two-level system cannot be inverted by light alone

Let the absorption, stimulated-emission and spontaneous-emission probability densities be p_{ab}, p_{st}, p_{sp}. With n photons in the mode, $p_{ab} = p_{st} = n\,(c/V)\sigma$ while $p_{sp} = (c/V)\sigma$ is independent of n. The populations obey

$$\frac{\mathrm{d}N_2}{\mathrm{d}t} = p_{ab}N_1 - (p_{st} + p_{sp})N_2.$$

At steady state $\mathrm{d}N_2/\mathrm{d}t = 0$, so

$$\frac{N_2}{N_1} = \frac{p_{ab}}{p_{st} + p_{sp}} = \frac{n}{n+1} < 1.$$

No matter how intense the light ($n \to \infty$ gives $N_2/N_1 \to 1$), you can at best equalise the populations. *A genuine inversion requires auxiliary levels and pumping* — this is why every real laser is a three- or four-level system.

The gain coefficient

Consider a thin slab of length $\mathrm{d}z$ and unit area. The net number of photons added per unit time per unit volume by stimulated processes is NW_i, where $N = N_2 - N_1$. The photon-flux density therefore increases across the slab by $\mathrm{d}\phi = NW_i\,\mathrm{d}z$. Writing $W_i = \phi\sigma(\nu)$,

$$\frac{\mathrm{d}\phi}{\mathrm{d}z} = N\sigma(\nu)\,\phi \equiv \gamma(\nu)\,\phi \quad \Longrightarrow \quad \phi(z) = \phi(0)\,\mathrm{e}^{\gamma(\nu)z}.$$

The growth is exponential because every emitted photon goes on to stimulate further emission. If $N < 0$ the exponent is negative and the medium attenuates — a medium in thermal equilibrium can never amplify.

Steady-state inversion from the rate equations

With pumping rates R_1, R_2 and level lifetimes τ_1, τ_2 (and τ_{21} for the $2 \to 1$ channel), the populations obey

$$\frac{\mathrm{d}N_2}{\mathrm{d}t} = R_2 - \frac{N_2}{\tau_2}, \qquad \frac{\mathrm{d}N_1}{\mathrm{d}t} = -R_1 - \frac{N_1}{\tau_1} + \frac{N_2}{\tau_{21}}.$$

Setting both derivatives to zero gives $N_2 = R_2\tau_2$ and $N_1 = \tau_1(N_2/\tau_{21} - R_1)$, so the inversion is

$$N_0 = N_2 - N_1 = R_2\tau_2\left(1 - \frac{\tau_1}{\tau_{21}}\right) + R_1\tau_1.$$

For an ideal four-level system $\tau_1 \ll t_{sp}$ and $\tau_2 \approx t_{sp}$, which collapses this to $N_0 \approx R_2 t_{sp}$. *Physical reading:* you want the upper level pumped hard and long-lived, and the lower level to empty instantly so it never blocks the transition.

Gain saturation

Adding stimulated terms $\pm N W_i$ to the rate equations and re-solving at steady state gives

$$N = \frac{N_0}{1+\tau_s W_i}, \qquad \tau_s = \tau_2 + \tau_1\left(1 - \frac{\tau_2}{\tau_{21}}\right) > 0.$$

Since $W_i = \phi\sigma$ and $\gamma = N\sigma$,

$$\gamma(\nu) = \frac{\gamma_0(\nu)}{1+\phi/\phi_s(\nu)}, \qquad \phi_s(\nu) = \frac{1}{\tau_s \sigma(\nu)}.$$

As the field builds up, it eats into the very inversion that produces it: the gain *saturates*. This negative feedback is what stabilises a laser.

Threshold and the clamped steady state

Oscillation needs the round-trip gain to beat the round-trip loss: $\gamma_0(\nu) > \alpha_r$, i.e. $N_0 > N_t \equiv \alpha_r/\sigma(\nu)$. Once oscillating, the photon flux grows until the *saturated* gain has fallen to exactly the loss:

$$\frac{\gamma_0(\nu)}{1+\phi/\phi_s(\nu)} = \alpha_r \quad \Longrightarrow \quad \phi = \phi_s(\nu)\left(\frac{\gamma_0(\nu)}{\alpha_r} - 1\right) = \phi_s(\nu)\left(\frac{N_0}{N_t} - 1\right).$$

This is *gain clamping*: above threshold the gain stays pinned at α_r and any extra pump power emerges as output light, so the output rises linearly with pump rate.

The Schawlow–Townes linewidth

Even a perfect single-mode laser has a finite linewidth because each spontaneous-emission event randomly kicks the phase of the intracavity field. Writing the field as $E = \sqrt{N_{\text{ph}}}\,\mathrm{e}^{\mathrm{i}\varphi}$, a single spontaneous photon adds a unit-amplitude term at random phase, giving a phase kick $\delta\varphi \sim 1/\sqrt{N_{\text{ph}}}$ and hence a variance $\delta\varphi^2 \sim 1/N_{\text{ph}}$. These kicks occur at the rate $1/\tau_p$, so the phase *diffuses*, and the resulting spectral width is

$$\Delta\nu_l \sim \frac{1}{\tau_p N_{\text{ph}}}.$$

Both high power (large N_{ph}) and high cavity Q (large τ_p) narrow the line. In practice technical noise — vibrations, temperature drift — dominates long before this quantum floor is reached.

Mode-locking produces a pulse train

If $M = 2S+1$ longitudinal modes, equally spaced by ν_F, are forced to have equal amplitudes A and locked phases, the total field is a geometric sum:

$$U(t,z) = \sum_{q=-S}^{S} A\,\mathrm{e}^{\mathrm{i}2\pi(\nu_0+q\nu_F)(t-z/c)}.$$

Summing the geometric series and taking the modulus squared gives the intensity quoted in the Core Equations, $I \propto \sin^2(M\pi t'/T_F)/\sin^2(\pi t'/T_F)$ with $t' = t - z/c$ — a train of pulses repeating every $T_F = 1/\nu_F$. Each pulse has duration $\tau_{\text{pulse}} = T_F/M = 1/\Delta\nu$ (the more modes locked, the shorter the pulse) and peak intensity $I_p = M\bar{I}$ (the energy of all M modes is concentrated into a brief spike). This is the time–bandwidth relation in action: a broad locked spectrum gives a short pulse.

5.5 Diagram Analysis

The energy-level diagrams distinguish absorption (upward, photon destroyed), stimulated emission (downward, clone photon created) and spontaneous emission (downward, photon into a random mode). The gain-versus-frequency plots show a Lorentzian profile of width $\Delta\nu$ centred on ν_0; the laser oscillates only where this curve exceeds the (flat) loss line, defining the oscillation bandwidth B. The longitudinal-mode comb is overlaid on the gain curve: only the discrete ν_q that fall under the gain-above-loss band can lase. For a homogeneously broadened medium the slides show the gain curve being pulled *uniformly* downward as it saturates until only the single mode nearest ν_0 survives; for an inhomogeneous medium they show discrete "holes" burned into the gain profile at each oscillating mode (spectral hole burning), allowing several modes to coexist. The mode-locking figure shows many random-phase modes producing noisy continuous output, versus phase-locked modes producing a clean periodic pulse train.

5.6 Example Problems

Longitudinal modes of a gas laser — Example Sheet 3, Question 1. For an Ar^+ laser this question works through the free spectral range $\nu_F = c/2d$, the number of longitudinal modes that fit under the gain curve ($M \approx B/\nu_F$, where B is the band over which gain exceeds loss), and the cavity length short enough for single-mode operation ($\nu_F \geq B$). The physical message: gas lasers are naturally multimode because the Doppler-broadened gain bandwidth spans many cavity modes, and single-mode operation means making the cavity short enough that only one mode fits under the gain curve.

Resonator loss and photon lifetime — Example Sheet 2, Question 8. For a ruby laser rod with given mirror reflectances this asks for the resonator loss coefficient $\alpha_r = (2d)^{-1}\ln(1/R_1R_2)$ and the photon lifetime $\tau_p = 1/(c\alpha_r)$. Both formulas are in the Core Equations for this Part; τ_p is the time a photon survives in the cavity before escaping, and it sets the threshold inversion ($N_t \propto 1/\tau_p$) and the Schawlow–Townes linewidth.

Common Pitfalls & Takeaways

1. "Stimulated emission alone can build an inversion." It cannot: in a pure two-level system $N_2/N_1 = n/(n+1) < 1$ for any photon number. Inversion is a *pumping* phenomenon requiring extra levels.
2. "More pumping means more gain." Only below threshold. Above threshold the gain is *clamped* at the loss value; extra pump power increases the output *photon flux*, not the gain coefficient.
3. Confusing the resonator linewidth with the gain linewidth. The gain bandwidth $\Delta\nu$ comes from the atomic transition; the cavity-mode spacing ν_F comes from the resonator geometry; the cold-cavity mode width $\delta\nu = \nu_F/\mathcal{F}$ comes from the resonator finesse. All three are different quantities and all three appear in laser physics — keep them distinct.
Takeaway: a laser is amplifier + pump + feedback. Threshold is "gain beats loss"; steady state is "saturated gain = loss" (clamping); the output spectrum is the joint verdict of the gain curve (which frequencies *can* oscillate) and the resonator modes (which frequencies *are allowed*).

6 Part V — Guided-Wave Optics (Lectures 13–14)

6.1 Course Overview for this Part

Free-space optics is bulky and fragile: mirrors and lenses take space, and beams are easily blocked or scattered. **Guided-wave optics** instead confines light inside dielectric conduits — waveguides and optical fibres — which form the wiring of photonic integrated circuits and the backbone of global telecommunications. This part builds the theory from the simplest model (light bouncing between two mirrors) upward: the *self-consistency condition* that quantises the allowed bounce angles into discrete **modes**, the modification of that condition for dielectric (total-internal-reflection) waveguides, the properties of optical fibres and their dispersion, two-dimensional waveguides, and how power is coupled into a guide and exchanged between neighbouring guides (directional couplers).

6.2 Key Concepts & Definitions

- **Waveguide mode:** a field distribution that keeps the same transverse profile and polarisation at every point along the guide — only an overall phase $\mathrm{e}^{-\mathrm{i}\beta z}$ changes.
- **Self-consistency condition:** the requirement that a wave reproduce itself after two reflections; it quantises the bounce angle θ_m and hence the mode index m.
- **Propagation constant β:** the z-component of the wavevector, $\beta = k\cos\theta$; higher-order modes have smaller β.
- **Cutoff:** for a mirror waveguide, the longest wavelength (lowest frequency) that can be guided, $\lambda_c = 2d$. A dielectric waveguide has *no* overall cutoff — its fundamental mode is always guided — but each higher mode has its own cutoff.
- **Numerical aperture NA:** $\sqrt{n_1^2 - n_2^2}$ for a dielectric guide; the sine of the acceptance angle for rays entering from outside.
- **Evanescent field:** in a dielectric guide the mode does not vanish at the core boundary but leaks a decaying tail into the cladding.
- **Single-mode vs multimode:** a guide thin enough (or a wavelength long enough) supports only the fundamental mode; wider guides support many.
- **Modal dispersion:** in a multimode guide different modes travel at different group velocities, spreading a pulse — the main speed limit of multimode fibre. *Graded-index* fibres compensate this.
- **Material dispersion / Rayleigh scattering:** wavelength dependence of n, and $1/\lambda^4$ scattering off frozen-in density fluctuations — the two main loss/spreading mechanisms in fibre.
- **Directional coupler:** two guides placed close enough that their evanescent fields overlap, so power oscillates back and forth between them; at the coupling length $L_0 = \pi/2C$ power is fully transferred.

6.3 Core Equations

Mirror waveguide: modes

$$\sin\theta_m = m\,\frac{\lambda}{2d}, \quad m = 1, 2, \ldots \qquad k_{ym} = \frac{m\pi}{d}, \qquad \beta_m^2 = k^2 - \frac{m^2\pi^2}{d^2}, \qquad M = \left\lfloor \frac{2d}{\lambda} \right\rfloor .$$

θ_m is the bounce angle of mode m; d the mirror separation (m); $\lambda = \lambda_0/n$ the wavelength in the medium (m); $k = nk_0$ the wavenumber (rad m^{-1}); k_{ym} the transverse wavevector component; β_m the propagation constant (rad m^{-1}); M the number of guided modes. The cutoff wavelength is $\lambda_c = 2d$.

Dielectric waveguide: modes

$$2k_y d - 2\varphi_r = 2\pi m, \quad m = 0, 1, 2, \ldots \qquad \beta_m = n_1 k_0 \cos\theta_m, \qquad M \approx \frac{2d}{\lambda_0}\mathrm{NA}, \quad \mathrm{NA} = \sqrt{n_1^2 - n_2^2}.$$

φ_r is the phase shift on total internal reflection (the extra ingredient absent for a mirror); n_1, n_2 are the core and cladding indices; $\theta_c = \cos^{-1}(n_2/n_1)$ is the complement of the critical angle. The single-mode condition is $\nu < \nu_c$ with $\nu_c = (1/\mathrm{NA})(c_0/2d)$.

Dispersion relation and group velocity (mirror guide)

$$\beta_m = \frac{\omega}{c}\sqrt{1 - m^2\frac{\omega_c^2}{\omega^2}}, \qquad v_m = \frac{\mathrm{d}\omega}{\mathrm{d}\beta_m} = c\cos\theta_m = c\sqrt{1 - m^2\frac{\omega_c^2}{\omega^2}}.$$

$\omega_c = \pi c/d$ is the cutoff angular frequency; v_m the group velocity of mode m (m s^{-1}). More oblique (higher-m) modes travel slower because their zig-zag path is longer.

Fibre attenuation and waveguide coupling

$$\alpha = \frac{1}{L}\,10\log_{10}\frac{P(0)}{P(L)}, \qquad P_1(z) = P_1(0)\cos^2(Cz), \quad P_2(z) = P_1(0)\sin^2(Cz), \quad L_0 = \frac{\pi}{2C}.$$

α is the attenuation coefficient (dB m^{-1}); L the fibre length (m); $P(0), P(L)$ the input and output powers (W); C the coupling coefficient between two identical guides (m^{-1}); L_0 the coupling (transfer) length (m).

6.4 Important Derivations

The self-consistency condition for a mirror waveguide

Picture a TEM plane wave bouncing between two mirrors separated by d at angle θ. For the wave to be a *mode* it must reproduce itself after one round trip of two reflections. Geometry shows that the extra path of the twice-reflected wave relative to the original is $2d\sin\theta$. Each perfect-mirror reflection adds a phase π, so two reflections add 2π — a full cycle, contributing nothing net. The self-consistency requirement is therefore that the path-difference phase be a multiple of 2π:

$$\frac{2\pi}{\lambda}\,2d\sin\theta = 2\pi m, \quad m = 1, 2, \ldots \quad \Longrightarrow \quad \sin\theta_m = m\,\frac{\lambda}{2d}.$$

Only these discrete bounce angles survive — the continuum of rays collapses into a discrete set of modes. Since $\sin\theta_m \le 1$, the largest allowed m is $\lfloor 2d/\lambda \rfloor$, giving the mode count M and the cutoff $\lambda_c = 2d$.

Propagation constant and why higher modes are slower

A guided mode is the superposition of two plane waves with wavevectors $(0, \pm k_y, \beta)$. The longitudinal component is $\beta = k\cos\theta$, and using $\sin\theta_m = m\lambda/2d$ with $\cos^2\theta = 1 - \sin^2\theta$,

$$\beta_m^2 = k^2(1 - \sin^2\theta_m) = k^2 - \frac{m^2\pi^2}{d^2}.$$

Higher m means a larger transverse wavevector $k_{ym} = m\pi/d$ and hence a smaller β_m — the wave zig-zags more steeply, so it makes slower progress down the guide. Differentiating the dispersion relation gives the group velocity $v_m = c\cos\theta_m$: the steeper the bounce, the slower the energy travels. This spread of v_m across modes is *modal dispersion*.

The dielectric waveguide differs by one phase term

For a dielectric guide, total internal reflection replaces the mirror. TIR is still loss-free, but it imposes an extra angle-dependent phase shift φ_r on each reflection. The self-consistency condition becomes

$$\frac{2\pi}{\lambda}\,2d\sin\theta - 2\varphi_r = 2\pi m, \quad m = 0, 1, 2, \dots$$

Two consequences follow. First, because φ_r varies smoothly with θ, the allowed angles are no longer evenly spaced. Second — and more importantly — the fundamental mode $m = 0$ is *always* a solution, so a dielectric waveguide has **no overall cutoff**: it guides at least one mode at any wavelength. (Each *higher* mode still has its own cutoff.) Counting solutions up to the critical angle gives $M \approx (2d/\lambda_0)\,\mathrm{NA}$.

Power exchange in a directional coupler

When two identical single-mode guides run close together, their evanescent tails overlap and the mode amplitudes a_1, a_2 obey the coupled equations

$$\frac{\mathrm{d}a_1}{\mathrm{d}z} = -\mathrm{i}C\,a_2, \qquad \frac{\mathrm{d}a_2}{\mathrm{d}z} = -\mathrm{i}C\,a_1$$

(for the phase-matched case $\beta_1 = \beta_2$). Differentiating the first and substituting the second gives $\mathrm{d}^2a_1/\mathrm{d}z^2 = -C^2 a_1$, a simple harmonic equation. With all power launched into guide 1, $a_1(0) = 1,\ a_2(0) = 0$, the solution is $a_1 = \cos(Cz),\ a_2 = -\mathrm{i}\sin(Cz)$, so the powers oscillate:

$$P_1(z) = P_1(0)\cos^2(Cz), \qquad P_2(z) = P_1(0)\sin^2(Cz).$$

At $z = L_0 = \pi/2C$ all the power has crossed into guide 2; at $L_0/2$ the split is 50:50 — the device is a 3 dB coupler, the integrated-optics equivalent of a beam splitter.

6.5 Diagram Analysis

The mode-profile diagrams show standing-wave patterns $u_m(y)$ across the guide: cosines for odd m (symmetric) and sines for even m (antisymmetric), with m nodes. For the *mirror* guide the field goes exactly to zero at the walls; for the *dielectric* guide it instead decays into an exponential evanescent tail in the cladding — the higher the mode, the deeper this tail penetrates. The fibre attenuation-versus-wavelength plot shows a deep low-loss window around 1.3–1.55 μm, flanked by ultraviolet electronic absorption (rising at short λ), mid-infrared vibrational absorption (rising at long λ), and the $1/\lambda^4$ Rayleigh scattering tail. The step-index, single-mode and graded-index fibre diagrams illustrate how a parabolic index profile equalises mode travel-times: rays that wander further from the axis spend their extra path in regions of lower index (higher speed), so all modes arrive together. The directional-coupler figure shows power sloshing sinusoidally between two adjacent guides.

6.6 Example Problem

Power carried by waveguide modes — Example Sheet 3, Question 3. This question shows that the $+z$ power carried by a single TE mode is $P_m = (|a_m|^2/2\eta)\cos\theta_m$, and — the important part — that for a multimode field the total power is simply $P = \sum_m P_m$, with *no* interference cross-terms. The mechanism is mode orthogonality: $\int u_m u_l \, dy = \delta_{ml}$ kills every cross-term in the Poynting-vector integral. This is precisely why a waveguide mode behaves as an independent "channel": power launched into one mode stays there and adds linearly with the others.

Common Pitfalls & Takeaways

1. "A dielectric waveguide has a cutoff like a mirror waveguide." It does not have an *overall* cutoff — the fundamental $m = 0$ mode propagates at any wavelength. What *is* true is that each *higher-order* mode has its own cutoff, and below a certain frequency only the fundamental survives (single-mode operation).
2. Confusing modal and material dispersion. Modal dispersion is a *geometric* effect — different modes take different-length zig-zag paths — and it afflicts multimode guides even at a single wavelength. Material dispersion is the wavelength dependence of n and afflicts even a single-mode guide carrying a finite-bandwidth pulse. Graded-index profiles fix the former, not the latter.
3. Forgetting the TIR phase shift. The single change from the mirror to the dielectric waveguide is the φ_r term in the self-consistency condition — but it is responsible for the uneven mode spacing, the absence of a cutoff, and the evanescent field. Do not drop it.
Takeaway: confinement quantises the continuum of rays into a discrete, orthogonal set of modes via one simple idea — the wave must reproduce itself after a round trip. Everything else (mode count, dispersion, coupling) follows from that condition.

7 Part VI — Optical Resonators (Lectures 15–16)

7.1 Course Overview for this Part

An optical resonator is the optical analogue of an electronic resonant circuit: it confines and stores light at discrete resonance frequencies set by its geometry. This part treats the two complementary descriptions of a resonator. The *ray* picture asks when a ray stays trapped after many round trips — yielding the **stability condition** for spherical-mirror resonators. The *wave* picture asks which field distributions reproduce themselves — yielding the resonance frequencies and showing that the modes of a stable spherical-mirror resonator are exactly the **Hermite–Gaussian beams**. Along the way we quantify the resonator by its **free spectral range**, **finesse**, **photon lifetime** and **quality factor** Q.

7.2 Key Concepts & Definitions

- **Fabry–Perot (planar) resonator:** two parallel flat mirrors; the simplest resonator. Highly sensitive to misalignment.
- **Resonance condition:** the round-trip phase must be a multiple of 2π; only then does an infinitesimal seed wave build up to finite intensity.
- **Free spectral range ν_F:** the frequency spacing between adjacent longitudinal resonances, $\nu_F = c/2d$.
- **Finesse $\mathcal{F}$:** the ratio of the free spectral range to the resonance linewidth, $\mathcal{F} = \nu_F/\delta\nu$; set by the mirror reflectivity and the losses.

- **Photon lifetime τ_p:** the $1/e$ decay time of the stored optical energy, $\tau_p = 1/(c\alpha_r)$; linked to the linewidth by $\delta\nu = 1/(2\pi\tau_p)$.
- **Quality factor Q:** $2\pi\times$(stored energy)/(energy lost per cycle); for an optical resonator $Q = 2\pi\nu_0\tau_p \approx \nu_0/\delta\nu$.
- **g-parameters:** $g_i = 1 + d/R_i$; the resonator is *stable* (rays confined) iff $0 \le g_1 g_2 \le 1$.
- **Confocal / concentric / planar resonators:** the special symmetric cases $d/|R| = 1, 2, 0$.
- **Hermite–Gaussian (HG) modes:** the family of beam solutions HG_{lm}, indexed by (l, m), that share the Gaussian wavefronts but have non-Gaussian intensity profiles; the modes of a spherical-mirror resonator.
- **Gouy phase and mode degeneracy:** the $(l + m + 1)\zeta(z)$ phase term shifts the resonance frequency of higher transverse modes; in the confocal resonator it makes whole families of modes degenerate.

7.3 Core Equations

Resonance frequencies, free spectral range, finesse

$$\nu_q = q\,\nu_F, \qquad \nu_F = \frac{c}{2d}, \qquad \mathcal{F} = \frac{\pi\sqrt{|r|}}{1-|r|} \approx \frac{\pi}{\alpha_r d}, \qquad \delta\nu \approx \frac{\nu_F}{\mathcal{F}}.$$

ν_q are the longitudinal resonance frequencies; ν_F the free spectral range (Hz); d the mirror separation (m); $|r|^2$ the round-trip intensity attenuation factor; $\mathcal{F}$ the finesse (dimensionless); α_r the loss coefficient (m^{-1}); $\delta\nu$ the resonance linewidth (Hz).

Photon lifetime and quality factor

$$\alpha_r = \alpha_s + \frac{1}{2d}\ln\frac{1}{R_1 R_2}, \qquad \tau_p = \frac{1}{c\alpha_r}, \qquad \delta\nu = \frac{1}{2\pi\tau_p}, \qquad Q = 2\pi\nu_0\tau_p \approx \frac{\nu_0}{\delta\nu} \approx \frac{\nu_0}{\nu_F}\mathcal{F}.$$

α_s is the distributed (absorption/scattering) loss (m^{-1}); R_1, R_2 the mirror reflectances; τ_p the photon lifetime (s); ν_0 the optical frequency (Hz); Q the quality factor (dimensionless). Because $\nu_0 \gg \nu_F$, always $Q \gg \mathcal{F}$.

Stability of a spherical-mirror resonator

$$g_1 = 1 + \frac{d}{R_1}, \qquad g_2 = 1 + \frac{d}{R_2}, \qquad \boxed{0 \le g_1 g_2 \le 1}$$

R_1, R_2 are the mirror radii of curvature (m; negative for concave); d the resonator length (m). For a symmetric resonator ($R_1 = R_2 = R$) this reduces to $0 \le d/|R| \le 2$.

Hermite–Gaussian modes and their resonance frequencies

$$U_{lm} = A_{lm}\frac{W_0}{W(z)} G_l\!\left(\frac{\sqrt{2}\,x}{W(z)}\right) G_m\!\left(\frac{\sqrt{2}\,y}{W(z)}\right) \exp\!\left[-\mathrm{i}kz - \mathrm{i}\frac{k\rho^2}{2R(z)} + \mathrm{i}(l+m+1)\zeta(z)\right]$$

$$\nu_{l,m,q} = q\,\nu_F + (l+m+1)\,\frac{\Delta\zeta}{\pi}\,\nu_F.$$

$G_l(u) = H_l(u)\,\mathrm{e}^{-u^2/2}$ is the Hermite–Gaussian function of order l; H_l a Hermite polynomial; (l, m) the transverse mode indices; q the longitudinal index; $\Delta\zeta = \zeta(z_2) - \zeta(z_1)$ the Gouy phase accumulated between the mirrors. HG_{00} is the ordinary Gaussian beam.

7.4 Important Derivations

Resonance frequencies of the planar resonator

The complex amplitude must satisfy the Helmholtz equation and vanish on both mirrors ($U = 0$ at $z = 0$ and $z = d$). The standing wave $U = A\sin(kz)$ already vanishes at $z = 0$; it also vanishes at $z = d$ only if $kd = q\pi$, so the wavenumber is quantised:

$$k_q = \frac{q\pi}{d} \quad\Longrightarrow\quad \nu_q = \frac{ck_q}{2\pi} = q\,\frac{c}{2d} = q\,\nu_F.$$

Equivalently (the travelling-wave picture): a seed wave returns after each round trip with phase $\phi = k\,2d$; the infinite sum of these phasors only builds to finite amplitude if they are all aligned, i.e. $\phi = 2\pi q$ — the same condition. The round-trip distance must be an integer number of wavelengths.

Finesse, photon lifetime and linewidth

With loss, successive round-trip phasors are reduced by a factor $h = |r|\,\mathrm{e}^{-\mathrm{i}\phi}$. Summing the geometric series, $U = U_0/(1-h)$, so the intracavity intensity is

$$I = |U|^2 = \frac{I_0}{|1-|r|\mathrm{e}^{-\mathrm{i}\phi}|^2} = \frac{I_{\max}}{1+(2\mathcal{F}/\pi)^2\sin^2(\phi/2)}, \qquad \mathcal{F} = \frac{\pi\sqrt{|r|}}{1-|r|}.$$

This is an Airy function: sharp transmission peaks at $\phi = 2\pi q$, with FWHM $\delta\phi \approx 2\pi/\mathcal{F}$. Converting phase to frequency via $\phi = 4\pi\nu d/c$ gives a frequency linewidth $\delta\nu \approx \nu_F/\mathcal{F}$. Now express this through energy decay: the round-trip loss is $|r|^2 = \mathrm{e}^{-2\alpha_r d}$, so α_r is the loss per unit length, $c\alpha_r$ the loss per unit time, and the stored energy decays as e^{-t/τ_p} with photon lifetime $\tau_p = 1/(c\alpha_r)$. A field decaying as $\mathrm{e}^{-t/2\tau_p}$ has a Lorentzian spectrum of width

$$\delta\nu = \frac{1}{2\pi\tau_p}.$$

The time–frequency uncertainty product is fixed: $\delta\nu\cdot\tau_p = 1/2\pi$. *Resonance broadening is simply the spectral signature of energy leaking out of the cavity.*

The stability condition from ray optics

A resonator is a *periodic* optical system: one round trip is two reflections and two propagations. The round-trip ray-transfer matrix is

$$M = \begin{pmatrix} 1 & 0 \\ -2/R_1 & 1 \end{pmatrix}\begin{pmatrix} 1 & d \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ -2/R_2 & 1 \end{pmatrix}\begin{pmatrix} 1 & d \\ 0 & 1 \end{pmatrix}.$$

After m round trips the ray height obeys the linear difference equation $y_{m+2} = 2b\,y_{m+1} - y_m$ with $b = (A+D)/2$ (the determinant is 1). Trying $y_m = y_0h^m$ gives the quadratic $h^2 - 2bh + 1 = 0$, so $h = b \pm \mathrm{i}\sqrt{1-b^2}$. If $|b| \le 1$ we can write $h = \mathrm{e}^{\pm\mathrm{i}\phi}$ with $\phi = \cos^{-1} b$ real, and the solution

$$y_m = y_{\max}\sin(m\phi + \phi_0)$$

is *bounded* — the ray oscillates about the axis forever and is confined. If $|b| > 1$ then h is real and one root exceeds 1: the ray height grows geometrically and the ray walks out of the resonator. The stability condition is therefore $|b| \le 1$. Evaluating b from the matrix product and introducing $g_i = 1 + d/R_i$ gives the compact form

$$b = 2g_1g_2 - 1, \qquad |b| \le 1 \iff 0 \le g_1g_2 \le 1.$$

Gaussian beams are the modes of a stable resonator

A Gaussian beam reflecting off a spherical mirror retraces itself *iff* its wavefront curvature there matches the mirror's. So the beam that "fits" a resonator is the one whose wavefront radii $R(z_1), R(z_2)$ at the mirror planes equal R_1, R_2. Imposing $R(z_1) = R_1$ and $R(z_2) = -R_2$ (the sign convention reflects that $R(z) > 0$ for $z > 0$) gives three equations whose solution fixes the beam-centre position and the Rayleigh range:

$$z_1 = -\frac{d(R_2 + d)}{R_1 + R_2 + 2d}, \qquad z_0^2 = -\frac{d(R_1 + d)(R_2 + d)(R_1 + R_2 + d)}{(R_1 + R_2 + 2d)^2}.$$

For the beam to be a genuine confined Gaussian, z_0^2 must be *positive*. Working through the algebra, $z_0^2 > 0$ is equivalent to

$$0 \le \left(1 + \frac{d}{R_1}\right)\left(1 + \frac{d}{R_2}\right) \le 1$$

— *exactly the ray-optics stability condition.* The two pictures agree: a resonator confines rays precisely when it can support a confined Gaussian mode.

Transverse mode frequencies and the confocal degeneracy

The on-axis phase of an HG_{lm} beam is $\varphi(0, z) = kz - (l + m + 1)\zeta(z)$. For the beam to retrace itself the round-trip phase change must be a multiple of 2π:

$$2kd - 2(l + m + 1)\,\Delta\zeta = 2\pi q \quad \Longrightarrow \quad \nu_{l,m,q} = q\,\nu_F + (l + m + 1)\,\frac{\Delta\zeta}{\pi}\,\nu_F.$$

The longitudinal spacing ν_F is unchanged, but transverse modes are shifted by the Gouy term. For the *symmetric confocal* resonator $d = |R|$, so $z_1 = -d/2$, $z_0 = d/2$, giving $\zeta(z_1) = \arctan(-1) = -\pi/4$, $\zeta(z_2) = +\pi/4$, hence $\Delta\zeta = \pi/2$ and

$$\nu_{l,m,q} = q\,\nu_F + (l + m + 1)\,\frac{\nu_F}{2}.$$

Modes with the same parity of $l + m$ are then *degenerate*: all even-$(l + m)$ modes pile up at one set of frequencies, all odd-$(l + m)$ modes at another (shifted by $\nu_F/2$). This robustness — many transverse modes resonating at the same frequency — is one reason the confocal geometry is so widely used.

7.5 Diagram Analysis

The stability diagram plots g_2 against g_1: the stable region is the area between the axes and the hyperbola $g_1 g_2 = 1$, occupying the first and third quadrants. Symmetric resonators lie on the diagonal $g_2 = g_1$; the planar ($g = 1$), confocal ($g = 0$) and concentric ($g = -1$) points sit at the corners and centre of the stable range, with confocal safely in the middle. The Airy-function plots of intracavity intensity versus frequency show that higher finesse produces sharper, more widely contrasted transmission peaks. The Hermite–Gaussian intensity patterns show HG_{00} as a single Gaussian spot, HG_{10} as two lobes, HG_{11} as a 2×2 array, and so on — the number of dark nodal lines equals the mode index in each direction, and all patterns scale with the common width $W(z)$.

7.6 Example Problem

Anatomy of a confocal resonator — Example Sheet 3, Question 8. For a symmetric confocal cavity this question works through (a) the mirror radii ($d = |R|$ by definition of confocal), (b) the waist of the fundamental mode (the waist sits at the centre with $z_0 = d/2$, so $W_0 = \sqrt{\lambda_0 z_0/\pi}$), and (c) the loss coefficient $\alpha_r = (2d)^{-1}\ln(1/R_1R_2)$ with its associated photon lifetime and finesse. Every formula needed is in the Core Equations for this Part. The takeaway is that the confocal geometry confines a tightly focused mode and, with high-reflectivity mirrors, stores each photon for many round trips — a high-Q, alignment-robust cavity.

Common Pitfalls & Takeaways

1. Confusing the three "widths". The free spectral range $\nu_F = c/2d$ is the *spacing* of resonances; the linewidth $\delta\nu = \nu_F/\mathcal{F}$ is the *width* of each resonance; the finesse $\mathcal{F}$ is their *ratio*. They scale differently with mirror reflectivity — only $\delta\nu$ and $\mathcal{F}$ depend on it.
2. Sign conventions for mirror radii. A concave mirror has $R < 0$ in this course's convention. Getting the sign wrong flips the g-parameters and can wrongly classify a stable resonator as unstable. Check that a symmetric concave-mirror resonator gives $0 \le d/|R| \le 2$.
3. "The ray and wave pictures give different stability conditions." They give the *same* condition, $0 \le g_1 g_2 \le 1$. The ray picture derives it from bounded ray trajectories; the wave picture derives it from the requirement $z_0^2 > 0$ for a confined Gaussian mode. Their agreement is a key consistency check, not a coincidence.
Takeaway: a resonator is characterised by two numbers — the free spectral range (geometry) and the finesse (loss) — and its modes are the Hermite–Gaussian beams. Stability has a ray meaning (rays stay trapped) and an identical wave meaning (a Gaussian mode exists).

8 Part VII — Quantum Optics (Lectures 17–22)

8.1 Course Overview for this Part

The final part takes the last conceptual step: it **quantises the electromagnetic field itself**. A single field mode is shown to be mathematically a harmonic oscillator, whose quanta are **photons**. The energy eigenstates are the **Fock (number) states**; the states that behave most like classical light are the **coherent states**; and the interplay between the two is laid bare by their phase-space portraits and photon statistics. The field is then coupled to a **two-level atom**: classically driven, this gives **Rabi oscillations**, the **optical Bloch equations** and the **Bloch sphere**; with the field also quantised, it gives **cavity QED** and the **Jaynes–Cummings model**. Finally, photon-correlation experiments — **Hanbury Brown–Twiss** and **Hong–Ou–Mandel** — reveal genuinely non-classical features of light: antibunching and two-photon interference.

8.2 Key Concepts & Definitions

- **Field quantisation:** a single mode of the EM field has the same Hamiltonian as a unit-mass harmonic oscillator; the canonical "position" and "momentum" are the electric and magnetic field amplitudes.
- **Creation/annihilation operators $\hat{a}^\dagger, \hat{a}$:** raise and lower the photon number by one; obey $[\hat{a}, \hat{a}^\dagger] = 1$.
- **Fock (number) state $|n\rangle$:** the eigenstate of the number operator $\hat{n} = \hat{a}^\dagger\hat{a}$ with exactly n photons; an energy eigenstate, but with completely random phase and zero mean field.
- **Vacuum energy:** the $\frac{1}{2}\hbar\omega$ in $\hat{H} = \hbar\omega(\hat{n} + \frac{1}{2})$; the field has energy even with zero photons.

- **Quadrature operators $\hat{X}_1, \hat{X}_2$:** the dimensionless "in-phase" and "out-of-phase" field components, $\hat{X}_1 = \frac{1}{2}(\hat{a}+\hat{a}^\dagger)$, $\hat{X}_2 = \frac{1}{2\mathrm{i}}(\hat{a}-\hat{a}^\dagger)$; together they carry amplitude and phase information.
- **Coherent state $|\alpha\rangle$:** the eigenstate of the annihilation operator, $\hat{a}\,|\alpha\rangle = \alpha\,|\alpha\rangle$; has a Poissonian photon distribution, a non-zero mean field, and vacuum-level fluctuations independent of α. The quantum model of an ideal laser beam.
- **Husimi Q function:** a non-negative phase-space quasi-probability distribution, $Q(\alpha) = \langle\alpha|\,\hat{\rho}\,|\alpha\rangle/\pi$, used to visualise quantum states.
- **Rabi model:** a two-level atom driven by a classical monochromatic field; after the rotating-wave approximation the populations oscillate at the (generalised) Rabi frequency.
- **Rotating-wave approximation (RWA):** dropping the fast-counter-rotating terms $(\omega+\omega_0)$ that average to zero, keeping only the near-resonant $(\omega - \omega_0 = \Delta)$ terms.
- **Optical Bloch equations:** the density-matrix equations of motion for a two-level atom *including* spontaneous emission; they admit a steady state.
- **Bloch vector / Bloch sphere:** the geometric representation (u, v, w) of the atomic density matrix; coherent evolution is precession about an axis $(-\Omega_0, 0, -\Delta)$, decoherence pulls the vector inside the sphere.
- **Saturation parameter S:** measures how hard the transition is driven; the steady-state fluorescence is $\Gamma_{sc} = (\gamma/2)S/(1+S)$, saturating at $\gamma/2$.
- **Dressed states:** the eigenstates of the atom-plus-field Hamiltonian; the field "dresses" the bare atomic states. Their intensity-dependent energy shift underlies dipole traps and optical tweezers.
- **Jaynes–Cummings model:** a two-level atom coupled to a single *quantised* cavity mode; predicts quantum Rabi oscillations at $\Omega = 2g\sqrt{n+1}$.
- **Second-order coherence $g^{(2)}(\tau)$:** the normalised intensity (photon-number) correlation; $g^{(2)}(0) \geq 1$ for classical light, $g^{(2)}(0) < 1$ (antibunching) is a purely quantum signature.
- **Hong–Ou–Mandel effect:** two indistinguishable photons entering opposite ports of a beam splitter always leave together; the coincidence rate dips to zero — a test of photon indistinguishability.

8.3 Core Equations

Quantised single-mode field

$$\hat{H} = \frac{1}{2}(\hat{p}^2 + \omega^2\hat{q}^2) = \hbar\omega\left(\hat{a}^\dagger\hat{a} + \tfrac{1}{2}\right) = \hbar\omega\left(\hat{n} + \tfrac{1}{2}\right), \qquad [\hat{a}, \hat{a}^\dagger] = 1.$$

$$\hat{a}\,|n\rangle = \sqrt{n}\,|n-1\rangle\,, \quad \hat{a}^\dagger\,|n\rangle = \sqrt{n+1}\,|n+1\rangle\,, \quad |n\rangle = \frac{(\hat{a}^\dagger)^n}{\sqrt{n!}}\,|0\rangle\,.$$

$\hat{q}, \hat{p}$ are the canonical field quadratures; ω the mode angular frequency (rad s^{-1}); $\hbar$ the reduced Planck constant (J s); $\hat{n}$ the photon-number operator (dimensionless eigenvalues $n = 0, 1, 2, \ldots$). The field operator is $\hat{E}_x(z) = \mathcal{E}_0(\hat{a} + \hat{a}^\dagger)\sin(kz)$ with $\mathcal{E}_0 = \sqrt{\hbar\omega/\epsilon_0 V}$ the "field per photon" (V m^{-1}) and V the mode volume (m^3).

Coherent states

$$\hat{a}\,|\alpha\rangle = \alpha\,|\alpha\rangle\,, \qquad |\alpha\rangle = \mathrm{e}^{-|\alpha|^2/2}\sum_{n=0}^{\infty}\frac{\alpha^n}{\sqrt{n!}}\,|n\rangle\,, \qquad P_n = \mathrm{e}^{-\bar{n}}\frac{\bar{n}^{\,n}}{n!}, \qquad \bar{n} = |\alpha|^2, \quad \Delta n = \sqrt{\bar{n}}.$$

$\alpha \in \mathbb{C}$ is the coherent-state amplitude; P_n the probability of finding n photons (Poissonian); $\bar{n}$ the mean photon number; Δn its standard deviation. The fractional uncertainty $\Delta n/\bar{n} = 1/\sqrt{\bar{n}} \to 0$ as $\bar{n} \to \infty$.

Rabi model and Rabi oscillations

$$H = \frac{\hbar}{2}\begin{pmatrix} -\Delta & \Omega_0 \\ \Omega_0 & \Delta \end{pmatrix}, \qquad \Delta = \omega - \omega_0, \qquad \Omega_0 = \frac{d\,E_0}{\hbar}, \qquad \Omega = \sqrt{\Omega_0^2 + \Delta^2}.$$

$$|\tilde{c}_e(t)|^2 = \frac{\Omega_0^2}{\Omega^2}\sin^2\left(\frac{\Omega t}{2}\right) \xrightarrow{\Delta=0} \sin^2\left(\frac{\Omega_0 t}{2}\right).$$

Δ is the detuning ($\mathrm{rad\,s^{-1}}$); Ω_0 the resonant Rabi frequency; Ω the generalised Rabi frequency; d the dipole matrix element ($\mathrm{C\,m}$); E_0 the field amplitude ($\mathrm{V\,m^{-1}}$); $|\tilde{c}_e|^2$ the excited-state population.

Optical Bloch equations and steady state

$$\dot{\rho}_{eg} = -(\gamma/2 - \mathrm{i}\Delta)\rho_{eg} - \frac{\mathrm{i}\Omega_0}{2}w, \qquad \dot{w} = -\gamma(w+1) - \mathrm{i}\Omega_0(\rho_{eg} - \rho_{ge}),$$

$$w_{st} = -\frac{1}{1+S}, \qquad S = \frac{\Omega_0^2/2}{\Delta^2 + \gamma^2/4} = \frac{S_0}{1 + 4\Delta^2/\gamma^2}, \qquad \Gamma_{sc} = \frac{\gamma}{2}\frac{S}{1+S}.$$

$w = \rho_{ee} - \rho_{gg}$ is the population inversion; ρ_{eg} the coherence; γ the spontaneous-emission rate ($\mathrm{s^{-1}}$); S the saturation parameter (dimensionless); $S_0 = 2\Omega_0^2/\gamma^2$ the resonant saturation parameter, proportional to intensity ($S_0 = I/I_{sat}$); Γ_{sc} the photon scattering rate ($\mathrm{s^{-1}}$).

Jaynes–Cummings model

$$H_{JC} = \hbar\omega_0\,\hat{\sigma}^\dagger\hat{\sigma} + \hbar\omega\,\hat{a}^\dagger\hat{a} + \hbar g\,(\hat{\sigma}^\dagger\hat{a} + \hat{\sigma}\hat{a}^\dagger), \qquad \Omega_n = 2g\sqrt{n+1}.$$

$\hat{\sigma} = |g\rangle\langle e|$ is the atomic lowering operator; g the atom–cavity coupling constant ($\mathrm{rad\,s^{-1}}$); ω_0 the atomic transition frequency; ω the cavity-mode frequency; Ω_n the quantum Rabi frequency for n photons. The state pairs $|e,n\rangle \leftrightarrow |g,n+1\rangle$ are closed under H_{JC}.

Second-order coherence

$$g^{(2)}(\tau) = \frac{\left\langle \hat{E}^{(-)}(t)\hat{E}^{(-)}(t+\tau)\hat{E}^{(+)}(t+\tau)\hat{E}^{(+)}(t)\right\rangle}{\left\langle \hat{E}^{(-)}(t)\hat{E}^{(+)}(t)\right\rangle^2}, \qquad g^{(2)}_{cl}(0) = \frac{\langle I^2\rangle}{\langle I\rangle^2} \geq 1.$$

$g^{(2)}(\tau)$ is the normalised intensity correlation (dimensionless). For classical light $g^{(2)}(0) \geq 1$ (coherent light: $= 1$; thermal light: $= 2$, photon *bunching*); a single emitter gives $g^{(2)}(0) = 0$ (*antibunching*), impossible classically.

8.4 Important Derivations

A field mode is a harmonic oscillator

For a single cavity mode the classical field energy $H = \frac{1}{2}\int(\epsilon_0 E^2 + \mu_0^{-1} B^2)\,\mathrm{d}V$ can, after writing the fields in terms of a single time-dependent amplitude $q(t)$ (with $p = \dot{q}$), be cast in the form

$$H = \tfrac{1}{2}\big(p^2 + \omega^2 q^2\big)\,.$$

This is *exactly* the Hamiltonian of a unit-mass harmonic oscillator. We quantise by promoting q, p to operators with $[\hat{q}, \hat{p}] = \mathrm{i}\hbar$. Introducing

$$\hat{a} = \frac{1}{\sqrt{2\hbar\omega}}(\omega\hat{q} + \mathrm{i}\hat{p}), \qquad \hat{a}^\dagger = \frac{1}{\sqrt{2\hbar\omega}}(\omega\hat{q} - \mathrm{i}\hat{p}),$$

one checks $[\hat{a}, \hat{a}^\dagger] = 1$ and the Hamiltonian becomes $\hat{H} = \hbar\omega(\hat{a}^\dagger\hat{a} + \frac{1}{2})$. The eigenvalues are $E_n = \hbar\omega(n + \frac{1}{2})$ — the quantum n is the *photon number*, and the $\frac{1}{2}\hbar\omega$ is the *vacuum energy*, present even when $n = 0$.

Fock states have zero mean field but non-zero fluctuations

The field operator is $\hat{E}_x \propto \hat{a} + \hat{a}^\dagger$. Because $\hat{a}$ and $\hat{a}^\dagger$ each change the photon number by one,

$$\langle n|\,(\hat{a} + \hat{a}^\dagger)\,|n\rangle = \sqrt{n}\,\langle n|n-1\rangle + \sqrt{n+1}\,\langle n|n+1\rangle = 0.$$

So $\left\langle \hat{E}_x \right\rangle = 0$ for every Fock state — it has no definite phase. But the *second* moment is non-zero. Using $(\hat{a} + \hat{a}^\dagger)^2 = \hat{a}^2 + \hat{a}^{\dagger 2} + \hat{a}^\dagger\hat{a} + \hat{a}\hat{a}^\dagger$ and discarding the number-changing terms,

$$\langle n|\,(\hat{a} + \hat{a}^\dagger)^2\,|n\rangle = \langle n|\,(\hat{a}^\dagger\hat{a} + \hat{a}\hat{a}^\dagger)\,|n\rangle = \langle n|\,(2\hat{a}^\dagger\hat{a} + 1)\,|n\rangle = 2n + 1,$$

where we used $\hat{a}\hat{a}^\dagger = \hat{a}^\dagger\hat{a} + 1$. Thus $\left\langle \hat{E}_x^2 \right\rangle \propto (n + \frac{1}{2})$. The field "has energy" but its phase is completely random — a Fock state is profoundly non-classical, no matter how large n is.

The coherent-state photon distribution is Poissonian

By definition $\hat{a}\,|\alpha\rangle = \alpha\,|\alpha\rangle$. Expanding $|\alpha\rangle = \sum_n c_n\,|n\rangle$ and applying $\hat{a}$ gives the recursion $c_n = (\alpha/\sqrt{n})c_{n-1}$, so $c_n = (\alpha^n/\sqrt{n!})c_0$; normalisation fixes $c_0 = \mathrm{e}^{-|\alpha|^2/2}$. The photon-number probability is then

$$P_n = |\,\langle n|\alpha\rangle\,|^2 = \mathrm{e}^{-|\alpha|^2}\frac{|\alpha|^{2n}}{n!} = \mathrm{e}^{-\bar{n}}\frac{\bar{n}^{\,n}}{n!},$$

a Poisson distribution with $\bar{n} = |\alpha|^2$. For a Poisson distribution the variance equals the mean, $(\Delta n)^2 = \bar{n}$, so the fractional uncertainty is $\Delta n/\bar{n} = 1/\sqrt{\bar{n}}$, which $\to 0$ as $\bar{n} \to \infty$. Moreover the field expectation is non-zero: $\langle \hat{a} + \hat{a}^\dagger \rangle = \alpha + \alpha^* = 2\,\mathrm{Re}(\alpha)$. *A coherent state has a well-defined amplitude and phase that sharpen — relative to the mean — as it gets brighter, which is precisely why it models an ideal laser.*

From the Schrödinger equation to the Rabi model

Write the atomic state as $|\Psi\rangle = c_g\,\mathrm{e}^{-\mathrm{i}E_g t/\hbar}\,|g\rangle + c_e\,\mathrm{e}^{-\mathrm{i}E_e t/\hbar}\,|e\rangle$ and insert it into $\mathrm{i}\hbar\,\partial_t\,|\Psi\rangle = \hat{H}\,|\Psi\rangle$ with $\hat{H} = \hat{H}_0 - \hat{d}\cdot\mathbf{E}(t)$ and a classical field $E(t) = E_0\cos\omega t$. Projecting onto $\langle g|$ and $\langle e|$ and using $\langle e|\,\hat{d}\,|g\rangle = d$ gives

$$\dot{c}_g = \mathrm{i}\Omega_0\,\mathrm{e}^{-\mathrm{i}\omega_0 t}\cos(\omega t)\,c_e, \qquad \dot{c}_e = \mathrm{i}\Omega_0\,\mathrm{e}^{+\mathrm{i}\omega_0 t}\cos(\omega t)\,c_g,$$

with $\Omega_0 = dE_0/\hbar$. Writing $\cos\omega t = \frac{1}{2}(\mathrm{e}^{\mathrm{i}\omega t} + \mathrm{e}^{-\mathrm{i}\omega t})$ produces terms oscillating at $\omega - \omega_0 = \Delta$ (slow) and at $\omega + \omega_0$ (fast). The **rotating-wave approximation** drops the fast terms, since they average to zero over the timescale of interest. After moving to a frame rotating at $\Delta/2$, the equations become

$$\frac{\mathrm{d}}{\mathrm{d}t}\begin{pmatrix}\tilde{c}_g \\ \tilde{c}_e\end{pmatrix} = \frac{\mathrm{i}}{2}\begin{pmatrix}-\Delta & \Omega_0 \\ \Omega_0 & \Delta\end{pmatrix}\begin{pmatrix}\tilde{c}_g \\ \tilde{c}_e\end{pmatrix}.$$

On resonance ($\Delta = 0$), differentiating once more gives $\ddot{\tilde{c}}_g = -(\Omega_0^2/4)\tilde{c}_g$, whose solution with the atom starting in the ground state is

$$\tilde{c}_g = \cos(\Omega_0 t/2), \qquad \tilde{c}_e = \mathrm{i}\sin(\Omega_0 t/2),$$

so the excited-state population oscillates as $|\tilde{c}_e|^2 = \sin^2(\Omega_0 t/2)$ — undamped **Rabi oscillations**. Off resonance the oscillation frequency becomes the generalised Rabi frequency $\Omega = \sqrt{\Omega_0^2 + \Delta^2}$ and the amplitude drops to $\Omega_0^2/\Omega^2 < 1$: a detuned drive can never fully invert the atom.

Spontaneous emission and the optical Bloch equations

The coherent Rabi equations conserve purity, so they cannot describe decay. We switch to the density matrix $\hat{\rho}$, evolving by $\mathrm{i}\hbar\dot{\hat{\rho}} = [\hat{H}, \hat{\rho}]$, and then add spontaneous emission *phenomenologically*: a $+\gamma\rho_{ee}$ term feeding the ground-state population, a $-\gamma\rho_{ee}$ draining the excited state, and a $-\gamma/2$ damping of the coherences. The result is the **optical Bloch equations**. Introducing the inversion $w = \rho_{ee} - \rho_{gg}$ reduces the four equations to two, quoted in the Core Equations. Setting $\dot{\rho} = 0$ gives the steady state

$$w_{st} = -\frac{1}{1+S}, \qquad S = \frac{\Omega_0^2/2}{\Delta^2 + \gamma^2/4}.$$

The scattered-photon rate $\Gamma_{sc} = \gamma\rho_{ee} = (\gamma/2)(1+w_{st}) = (\gamma/2)\,S/(1+S)$. At low intensity ($S \ll 1$) it rises linearly with I (constant cross section); at high intensity it *saturates* at $\Gamma_{sc}^{\max} = \gamma/2$ — the atom cannot scatter faster than its decay allows. The lineshape is Lorentzian of width γ at low power, broadening to $\gamma' = \gamma\sqrt{1+S_0}$ at high power (*power broadening*).

The Jaynes–Cummings model and quantum Rabi oscillations

Now quantise the field too. The atom–field coupling in the dipole approximation is $H_{AF} = \hbar g(\hat{\sigma}+\hat{\sigma}^\dagger)(\hat{a}+\hat{a}^\dagger)$. Multiplying out gives four terms; the RWA discards the energy-non-conserving $\hat{\sigma}\hat{a}$ (atom de-excites *and* destroys a photon) and $\hat{\sigma}^\dagger\hat{a}^\dagger$ (atom excites *and* creates a photon), leaving the **Jaynes–Cummings** interaction $\hbar g(\hat{\sigma}^\dagger\hat{a}+\hat{\sigma}\hat{a}^\dagger)$ — "atom goes up, photon disappears" and its reverse. Crucially, H_{JC} only ever couples the pair $|e,n\rangle \leftrightarrow |g,n+1\rangle$ to each other and to nothing else. Within each such two-dimensional subspace the dynamics is again a Rabi oscillation, but now with a *photon-number-dependent* frequency

$$\Omega_n = 2g\sqrt{n+1}.$$

Even with the cavity in vacuum ($n = 0$) the rate is $2g$ — the atom undergoes *vacuum Rabi oscillations*, exchanging a single quantum with the empty cavity mode. For a coherent-state cavity field, the superposition of many n means many frequencies $\propto \sqrt{n+1}$ beat against each other, and the Rabi signal *collapses* (and later *revives*) — a purely quantum effect with no classical analogue.

Two photons on a beam splitter: the HOM effect

A lossless 50:50 beam splitter relates the input and output mode operators by $\hat{a}_2 = \frac{1}{\sqrt{2}}(\hat{a}_0 + \mathrm{i}\hat{a}_1)$, $\hat{a}_3 = \frac{1}{\sqrt{2}}(\mathrm{i}\hat{a}_0 + \hat{a}_1)$, which inverts to $\hat{a}_0^\dagger = \frac{1}{\sqrt{2}}(\hat{a}_2^\dagger + \mathrm{i}\hat{a}_3^\dagger)$, $\hat{a}_1^\dagger = \frac{1}{\sqrt{2}}(\mathrm{i}\hat{a}_2^\dagger + \hat{a}_3^\dagger)$. Sending one photon into each input port, the input state is $|1\rangle_0 |1\rangle_1 = \hat{a}_0^\dagger \hat{a}_1^\dagger |0\rangle$. Substituting,

$$\hat{a}_0^\dagger \hat{a}_1^\dagger = \tfrac{1}{2}(\hat{a}_2^\dagger + \mathrm{i}\hat{a}_3^\dagger)(\mathrm{i}\hat{a}_2^\dagger + \hat{a}_3^\dagger) = \tfrac{1}{2}\Big(\mathrm{i}\,\hat{a}_2^{\dagger 2} + \hat{a}_2^\dagger \hat{a}_3^\dagger - \hat{a}_3^\dagger \hat{a}_2^\dagger + \mathrm{i}\,\hat{a}_3^{\dagger 2}\Big)\,.$$

The two cross terms $\hat{a}_2^\dagger \hat{a}_3^\dagger$ and $\hat{a}_3^\dagger \hat{a}_2^\dagger$ are equal (the operators commute) and *cancel*. Only the "both photons in the same port" terms survive:

$$|1\rangle_0 |1\rangle_1 \xrightarrow{BS} \frac{\mathrm{i}}{\sqrt{2}}(|2\rangle_2 |0\rangle_3 + |0\rangle_2 |2\rangle_3)\,.$$

The two indistinguishable photons *always* leave together — the amplitude for them to split (one to each output) has destructively interfered to zero. In a coincidence measurement the count rate therefore drops to zero. Introducing a relative delay τ makes the photons distinguishable and restores coincidences, tracing out the **Hong–Ou–Mandel dip**; its depth and width measure the photons' indistinguishability.

8.5 Diagram Analysis

The harmonic-oscillator ladder diagram shows the evenly spaced Fock levels $E_n = \hbar\omega(n + \frac{1}{2})$, the lowest sitting at $\frac{1}{2}\hbar\omega$ above zero — the vacuum energy. The phase-space cartoons are central: the *vacuum* is a fuzzy disk at the origin; a *coherent state* is the *same* disk displaced to $\alpha = |\alpha|e^{\mathrm{i}\Theta}$ (hence "displaced vacuum"), with fixed-size fluctuations so that as $|\alpha|$ grows the angular spread $\Delta\Theta$ and the fractional amplitude spread shrink toward the classical limit; a *Fock state* is a ring (sharp radius $\sqrt{n}$, completely random phase). The corresponding Husimi Q-function plots make this quantitative — a Gaussian blob for coherent states, a radially symmetric ring for Fock states. The Bloch-sphere figures show coherent evolution as precession of the Bloch vector about the axis $(-\Omega_0, 0, -\Delta)$: on resonance the vector sweeps great circles through the poles (a π-pulse swaps ground and excited states, a $\pi/2$-pulse makes an equal superposition); with spontaneous emission the vector spirals *inside* the sphere toward the steady state (mixed states have $u^2 + v^2 + w^2 < 1$). The Jaynes–Cummings energy diagram shows the characteristic "ladder of doublets": each rung $|e, n\rangle \,/\, |g, n+1\rangle$ splits by $\Omega_n = 2g\sqrt{n+1}$. The $g^{(2)}(\tau)$ plots distinguish the three regimes: flat at 1 (coherent light), a bump up to 2 at $\tau = 0$ decaying over the coherence time (thermal light, *bunching*), and a dip to 0 at $\tau = 0$ (single emitter, *antibunching*). The HOM schematic shows the coincidence rate falling into a sharp dip as the photon delay is tuned through zero.

8.6 Example Problems

The field operator algebra — Example Sheet 4, Question 1. This question asks you to verify $[\hat{a}, \hat{a}^\dagger] = 1$ from the definitions of $\hat{a}, \hat{a}^\dagger$ in terms of $\hat{q}, \hat{p}$, and then to recast the single-mode Hamiltonian as $\hat{H} = \hbar\omega(\hat{a}^\dagger\hat{a} + \frac{1}{2})$. Both steps are carried out in full in the *Quantising a single field mode* derivation above. The point worth carrying away: the single commutator $[\hat{a}, \hat{a}^\dagger] = 1$ is the entire algebraic engine of the quantised field — it generates the photon-number ladder, the $\frac{1}{2}\hbar\omega$ vacuum energy, and every field fluctuation.

Resonant and detuned Rabi pulses — Example Sheet 4, Question 3. This question tracks the excited-state population of a two-level atom when a field is switched on for a time T. On resonance the population is the full-contrast oscillation $\sin^2(\Omega_0 T/2)$ — a π-pulse ($T = \pi/\Omega_0$) fully inverts the atom; off resonance the generalised Rabi frequency $\Omega = \sqrt{\Omega_0^2 + \Delta^2}$ makes the oscillation faster but caps its amplitude at $\Omega_0^2/\Omega^2 < 1$. The derivation is the *Rabi model* result above. The physical lesson: resonant pulses enable complete coherent state transfer (the basis of atomic clocks and qubit control), and uncontrolled detunings are a primary error source because they both speed up and cap the oscillation.

Common Pitfalls & Takeaways

1. "A Fock state with large n is basically classical light." No. However large n is, a Fock state has $\left\langle \hat{E} \right\rangle = 0$ and a completely random phase. The classical limit is approached by *coherent* states, not number states — they are two profoundly different things.
2. Forgetting the vacuum port of a beam splitter. The beam-splitter relations cannot be written consistently without the unused input mode $\hat{a}_0$. Even when "nothing" enters that port, the vacuum fluctuations are physically there and affect the photon statistics — this is why the HOM derivation needs both inputs.
3. Misreading $g^{(2)}(0)$. $g^{(2)}(0) = 1$ is coherent (laser-like) light, *not* "no correlation in an everyday sense"; $g^{(2)}(0) = 2$ is thermal light (bunching); $g^{(2)}(0) < 1$ is non-classical antibunching. The classical bound is $g^{(2)}(0) \geq 1$ — beating it is a definite signature of quantum light.
4. Applying the RWA blindly. The rotating-wave approximation is valid only when the detuning is small compared with the optical frequency, $|\Delta| \ll \omega, \omega_0$. It is what makes both the Rabi and Jaynes–Cummings models tractable, but it is an approximation, not an identity.
Takeaway: quantising the field replaces "the amplitude of a wave" with "the number of photons", and the price is an irreducible uncertainty between photon number and phase. Fock states have definite number and random phase; coherent states balance the two and recover classical behaviour in the bright limit; and genuinely non-classical light (antibunched, or HOM-interfering) has no classical description at all.

9 Synthesis — The Four Levels of Light

The course is held together by a single organising idea: optics can be described at four nested levels, each the limiting case of the one below it.

The hierarchy of descriptions

Ray optics ($\lambda \to 0$) — light is rays obeying geometric rules; the tool is the ABCD matrix. Captures imaging; misses diffraction and interference.

Wave optics — light is a scalar field obeying the Helmholtz equation; the tools are the Gaussian beam, the coherence function and the Fourier transform. Captures diffraction, interference and the diffraction limit; misses polarisation.

Electromagnetic optics — light is the full vector Maxwell field; this is the level at which the *material response* (linear susceptibility, and then the nonlinear $\chi^{(2)}, \chi^{(3)}$) enters, giving the refractive index, birefringence, harmonic generation and the electro-optic and Kerr effects.

Quantum optics — the field itself is quantised into photons; the tools are the creation/annihilation operators, Fock and coherent states, and the density matrix. Captures spontaneous emission, photon statistics, antibunching and two-photon interference — everything with no classical analogue.

Each part of this handout lives mainly at one level but borrows from the others. The laser (Part IV) needs *quantum* stimulated emission for its gain, *wave* optics for its resonator modes, and *ray* optics for its stability condition. The Gaussian beam introduced in wave optics (Part II) reappears unchanged as the resonator mode (Part VI) and as the spatial profile of the quantised cavity field (Part VII). The ABCD matrix of ray optics (Part I) propagates Gaussian beams via the ABCD law. Coherence (Part II) returns as the first-order coherence $g^{(1)}$ whose quantum partner $g^{(2)}$ distinguishes classical from non-classical light (Part VII). Modern photonics — gravitational-wave interferometers, optical frequency combs, integrated quantum photonic circuits, single-atom tweezers — draws on all four levels at once.

A unifying recurrence: self-consistency and round trips. Notice how often the same logical move appears. A waveguide mode is a field that *reproduces itself after a round trip* of two reflections (Part V). A resonator mode is a field that *reproduces itself after a round trip* between the mirrors (Part VI). A laser oscillates when the *round-trip* gain beats the *round-trip* loss and the *round-trip* phase is a multiple of 2π (Part IV). Phase matching is the condition that three waves stay in step *as they co-propagate* (Part III). In every case, discreteness — of modes, of resonance frequencies, of allowed processes — emerges from a self-consistency requirement imposed on a wave.

A unifying recurrence: conjugate uncertainty. Time and frequency ($\tau_c\,\Delta\nu_c \sim 1$, the coherence relation; $\delta\nu\,\tau_p = 1/2\pi$, the resonator linewidth; $\tau_{\text{pulse}} = 1/\Delta\nu$, the mode-locked pulse). Position and transverse wavevector (the diffraction limit; $2W_0 \cdot 2\theta_0 = 4\lambda/\pi$, the beam quality bound). Photon number and phase (the Fock state has definite n and random phase; the coherent state shares the uncertainty between them). The same Fourier-conjugate structure underlies all of them.

10 Synthesis: How the Four Levels Fit Together

It is worth stepping back at the end of the course to see the whole structure as a single nested hierarchy, since every part is a limiting case of the one below it.

The four levels of description

- **Ray optics** (Part I) is the $\lambda \to 0$ limit. Light is a set of rays; the ABCD matrix is the complete toolkit. No diffraction, no interference.
- **Wave optics** (Parts II, V, VI) restores the finite wavelength. Light is a scalar field obeying the Helmholtz equation; this brings Gaussian beams, diffraction, coherence, waveguide modes and resonator modes. The *same* ABCD matrix now propagates the q-parameter of a Gaussian beam — ray optics survives inside wave optics.
- **Electromagnetic optics** (Part III) restores the vector nature and the material response. Polarisation, birefringence, and the nonlinear terms $\chi^{(2)}, \chi^{(3)}$ live here; lasers (Part IV) need the medium's gain, which is electromagnetic light–matter interaction treated semi-classically.
- **Quantum optics** (Part VII) quantises the field itself. Photons, Fock and coherent states, spontaneous emission, cavity QED and photon correlations appear. The classical wave re-emerges as the bright-coherent-state limit.

Each arrow "$\to$" in ray $\to$ wave $\to$ electromagnetic $\to$ quantum *adds* physics; nothing from the previous level is discarded, it is contained as a limit.

10.1 Master Equation Reference

The single most useful equations of the course, collected for revision:

Topic	Key relation
Ray transfer	$\begin{pmatrix} y_2 \\ \theta_2 \end{pmatrix} = \begin{pmatrix} A & B \\ C & D \end{pmatrix} \begin{pmatrix} y_1 \\ \theta_1 \end{pmatrix}$, imaging $\frac{1}{s_o} + \frac{1}{s_i} = \frac{1}{f}$
Helmholtz equation	$\nabla^2 U + k^2 U = 0, \quad k = 2\pi/\lambda$
Gaussian beam	$W(z) = W_0\sqrt{1+(z/z_0)^2}, \quad z_0 = \pi W_0^2/\lambda, \quad q = z + \mathrm{i}z_0$, ABCD law $q_2 = \frac{Aq_1+B}{Cq_1+D}$
Coherence	$g(\tau) = G(\tau)/G(0), \quad \tau_c = \int \lvert g\rvert^2 \mathrm{d}\tau, \quad \Delta\nu_c = 1/\tau_c, \quad \mathcal{V} = \lvert g_{12}\rvert$ (equal intensities)
Fourier optics	far field $g \propto F(x/\lambda d,\, y/\lambda d)$; Airy radius $1.22\,\lambda d/D$; Abbe limit $\lambda/2\mathrm{NA}$
Nonlinear polarisation	$P = \epsilon_0(\chi E + \chi^{(2)}E^2 + \chi^{(3)}E^3 + \cdots)$; phase match $\mathbf{k}_1 + \mathbf{k}_2 = \mathbf{k}_3$
Electro-optic / Kerr	$n(E_0) = n_0 + \frac{\chi^{(2)}}{2n_0}E_0; \quad n(I) = n_0 + n_2 I$
Laser gain	$\gamma(\nu) = N\sigma(\nu), \quad \mathrm{d}\phi/\mathrm{d}z = \gamma\phi$, threshold $\gamma_0(\nu) > \alpha_r, \quad N_t = \alpha_r/\sigma$
Resonator	$\nu_F = c/2d, \quad F = \pi\sqrt{\lvert r\rvert}/(1-\lvert r\rvert), \quad \tau_p = 1/(c\alpha_r)$, stability $0 \le g_1 g_2 \le 1$
Waveguide modes	mirror: $\sin\theta_m = m\lambda/2d, \; M = \lceil 2d/\lambda \rceil$; dielectric: $M \approx (2d/\lambda_0)\mathrm{NA}$
Field quantisation	$\hat{H} = \hbar\omega(\hat{a}^\dagger\hat{a} + \frac{1}{2}), \quad [\hat{a}, \hat{a}^\dagger] = 1, \quad \hat{a}\lvert n\rangle = \sqrt{n}\,\lvert n-1\rangle$
Coherent state	$\hat{a}\lvert\alpha\rangle = \alpha\lvert\alpha\rangle, \quad \bar{n} = \lvert\alpha\rvert^2, \quad (\Delta n)^2 = \bar{n}, \quad \Delta n/\bar{n} = 1/\lvert\alpha\rvert$
Atom–light (Rabi)	$H = \frac{\hbar}{2}\begin{pmatrix} -\Delta & \Omega_0 \\ \Omega_0 & \Delta \end{pmatrix}, \quad \Omega = \sqrt{\Omega_0^2 + \Delta^2}, \quad \lvert c_e\rvert^2 = \frac{\Omega_0^2}{\Omega^2}\sin^2(\Omega t/2)$
Optical Bloch / steady state	$w_{\mathrm{st}} = -1/(1+S), \quad S = S_0/(1 + 4\Delta^2/\gamma^2), \quad \Gamma_{\mathrm{sc}} = \frac{\gamma}{2}\frac{S}{1+S}$
Photon statistics	$g^{(2)}(0) \ge 1$ (classical); $= 1$ coherent, $= 2$ thermal, < 1 antibunched

10.2 Final Word

The recurring lesson of the course is *economy of description*: a Gaussian beam is one complex number q; a partially coherent source is one function $g(\tau)$; a nonlinear crystal is one symmetry rule plus two conservation laws; a laser is gain versus loss; a resonator is free spectral range and finesse; a quantum state of light is a point (or a blob) in phase space. Wherever possible, identify the minimal set of parameters that controls the physics — the algebra then takes care of itself.

End of handout. Read it alongside the problem sheets: the example-sheet questions are flagged at the relevant points in the text rather than worked through here, so that this document and the problem sheets complement rather than duplicate each other. Working those questions is the best test of whether the ideas have landed.